

The Invisible Graveyard: Quantifying the Mortality Cost of FDA Efficacy Lag, 1962-2024

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i The Short Version

The 1962 Kefauver-Harris Amendments require a 8.2 years (95% CI: 4.85 years-11.5 years) efficacy delay after drugs are proven safe. This creates two distinct mortality costs:

1. **Historical deaths (1962-2024):** 102M deaths (95% CI: 36.9M deaths-214M deaths) people died waiting for approved drugs during their approval process - a **lower bound** excluding drugs never developed due to cost barriers
2. **Future timeline shift (under cascade assumption):** 416M deaths (95% CI: 225M deaths-630M deaths) additional deaths will occur because the entire disease eradication timeline is pushed back by 8.2 years (95% CI: 4.85 years-11.5 years)

The ratio: Type II errors (blocking effective drugs) cost 3.07k:1 (95% CI: 2.88k:1-3.12k:1) more lives than Type I errors (approving dangerous drugs) prevent.

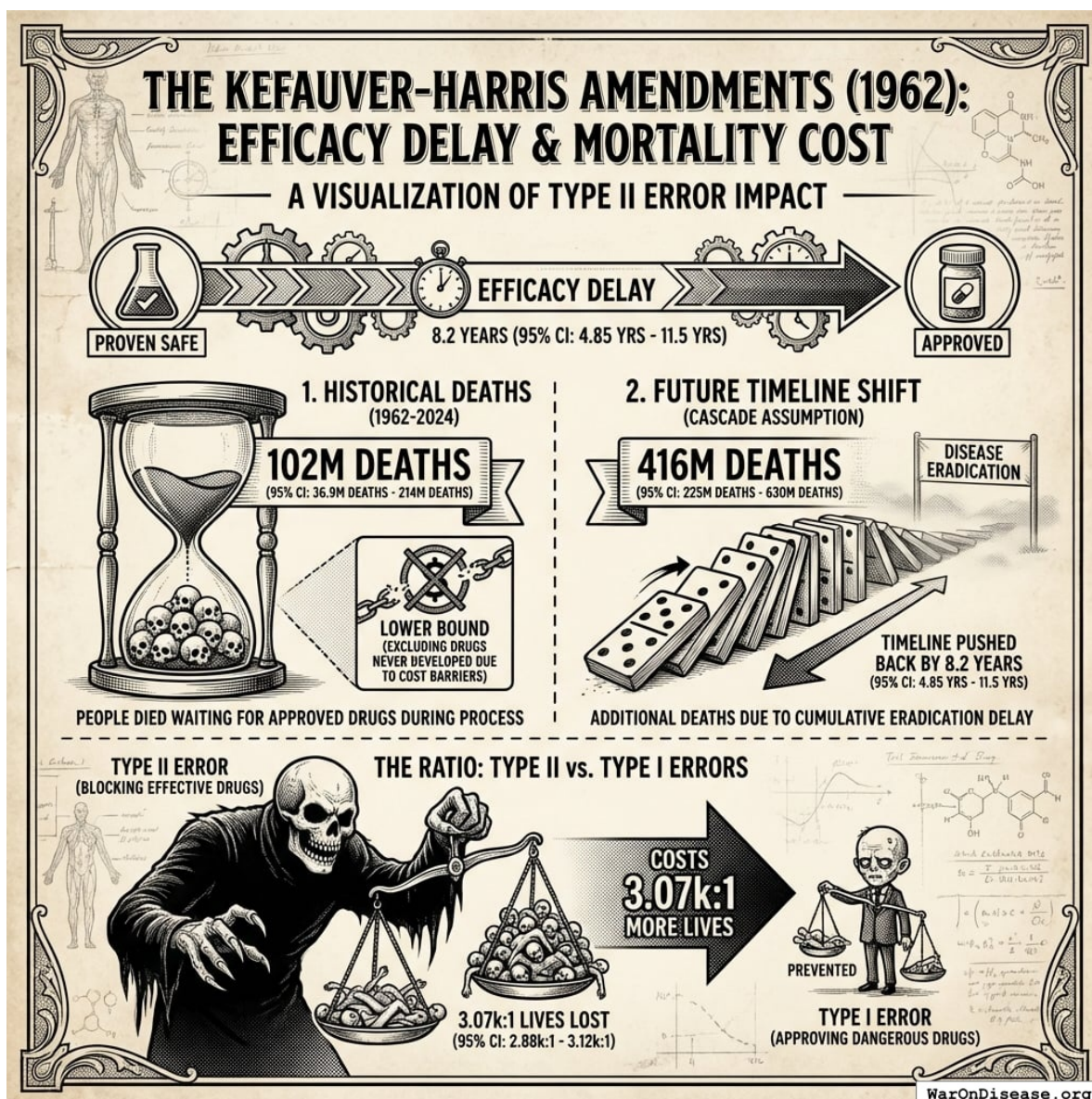


Figure 1: A comparison of historical and future mortality costs caused by drug efficacy delays, highlighting the 3,070:1 ratio between Type II and Type I error consequences.

1 Abstract

This study quantifies the cumulative mortality and morbidity costs associated with the **Unitary Pre-Market Approval (UPMA)** model mandated by the 1962 Kefauver-Harris Amendments. By enforcing efficacy testing prior to market entry, the current regulatory framework imposes an average “Efficacy Lag” of **8.2 years (95% CI: 4.85 years-11.5 years)** post-safety verification.

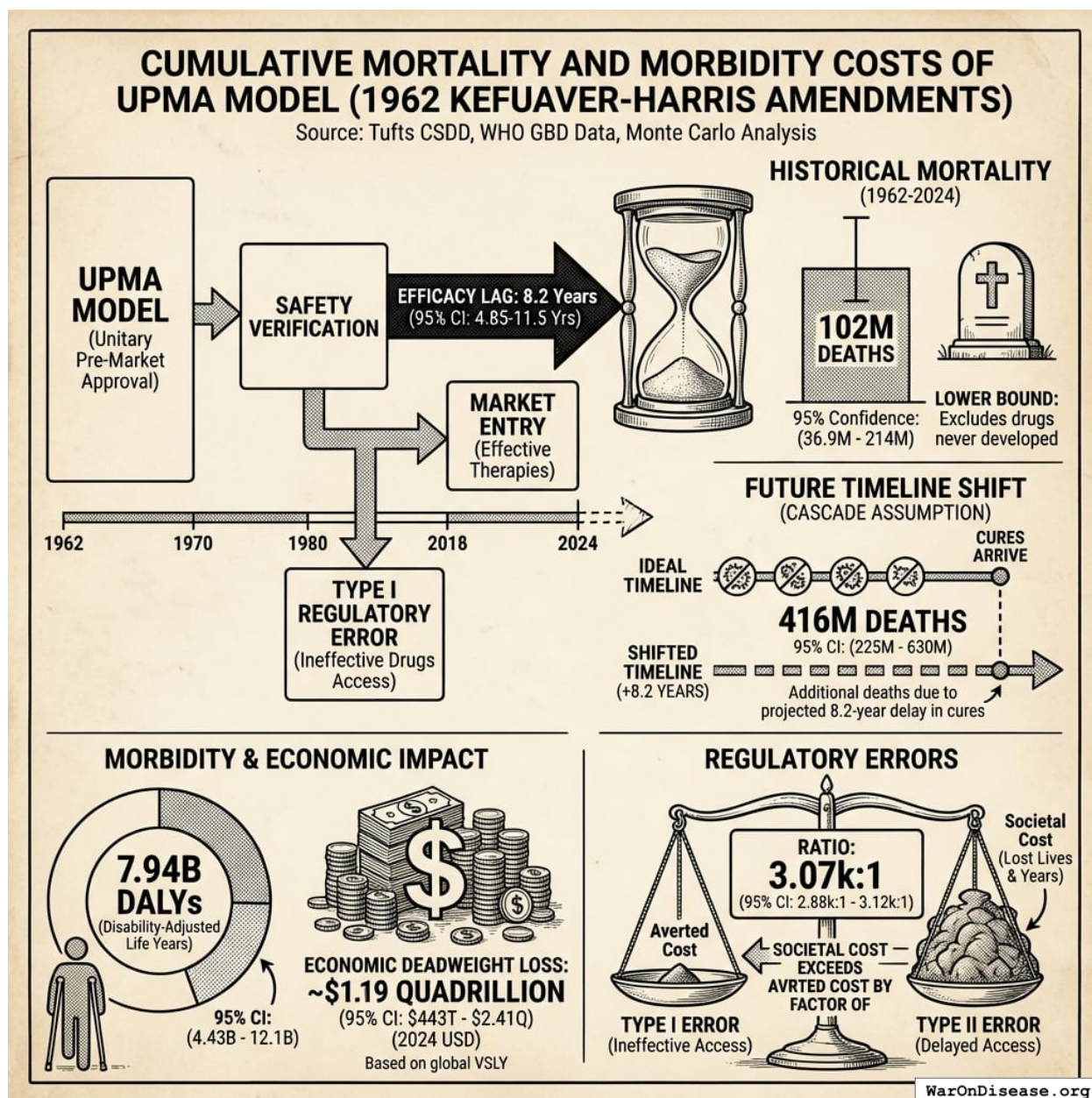


Figure 2: A comparative bar chart showing the scale of historical versus future mortality costs and the massive 3,070:1 ratio between Type II and Type I regulatory errors.

Using data from the Tufts Center for the Study of Drug Development (CSDD) and the WHO Global Burden of Disease (GBD) database, we estimate two distinct mortality costs:

1. **Historical mortality (1962-2024):** Approximately **102M deaths** (95% CI: 36.9M deaths-214M deaths) died waiting for approved drugs during their 8.2 years (95% CI: 4.85 years-11.5 years) approval delays. This is a **lower bound** - it excludes drugs never developed due to cost barriers.
2. **Future timeline shift (under cascade assumption):** An additional **416M deaths** (95% CI: 225M deaths-630M deaths) will eventually die because the entire disease eradication

timeline has been pushed back by 8.2 years (95% CI: 4.85 years-11.5 years). When cures finally arrive, they arrive 8.2 years (95% CI: 4.85 years-11.5 years) later than they would have without efficacy requirements. During that delay, people die.

Historical Deaths Calculation:

$$\begin{aligned}
 Deaths_{lag,total} &= Lives_{saved,annual} \times T_{lag} \\
 &= 12.4M \times 8.2 \\
 &= 102M
 \end{aligned}$$

$$\text{where } Lives_{saved,annual} = \frac{LY_{saved,annual}}{T_{ext}} = \frac{149M}{12} = 12.4M$$

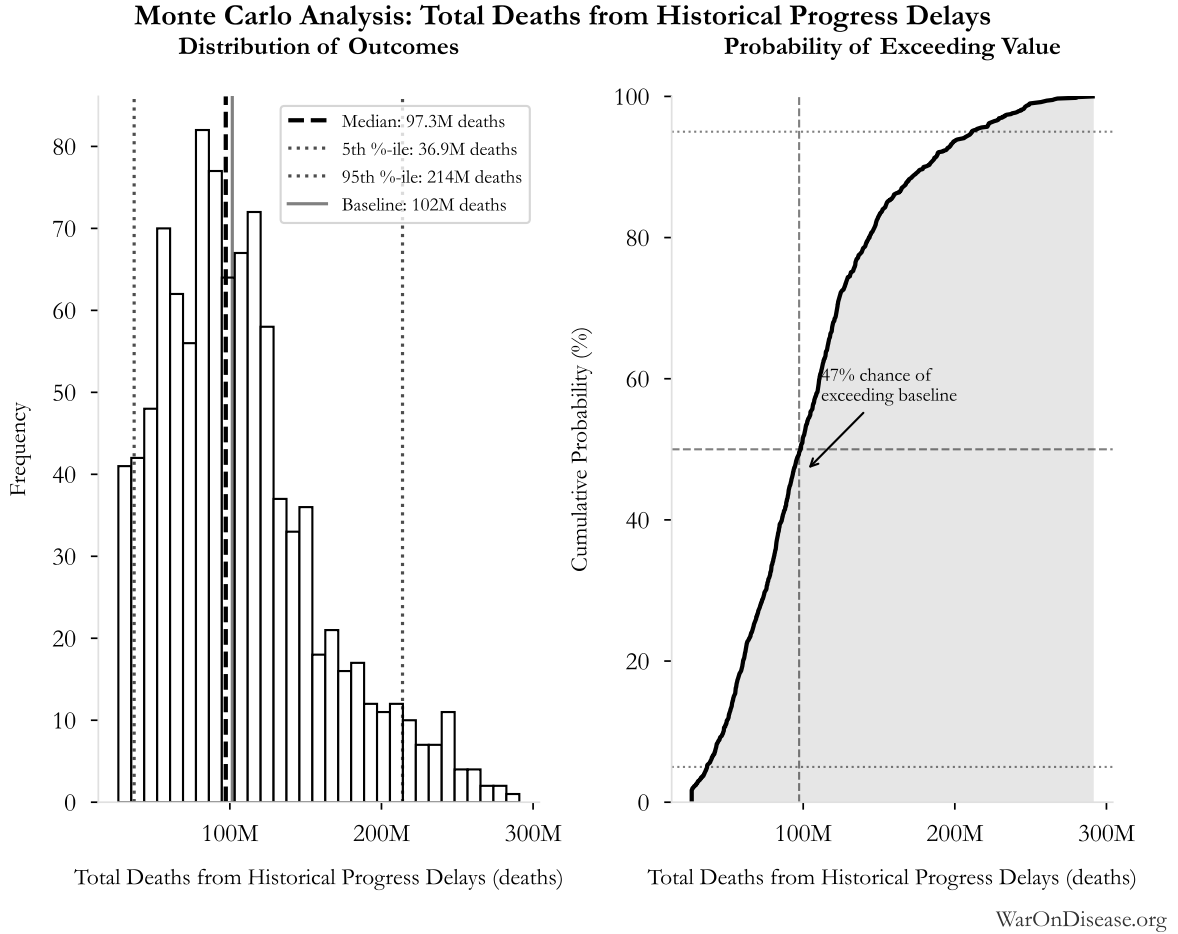
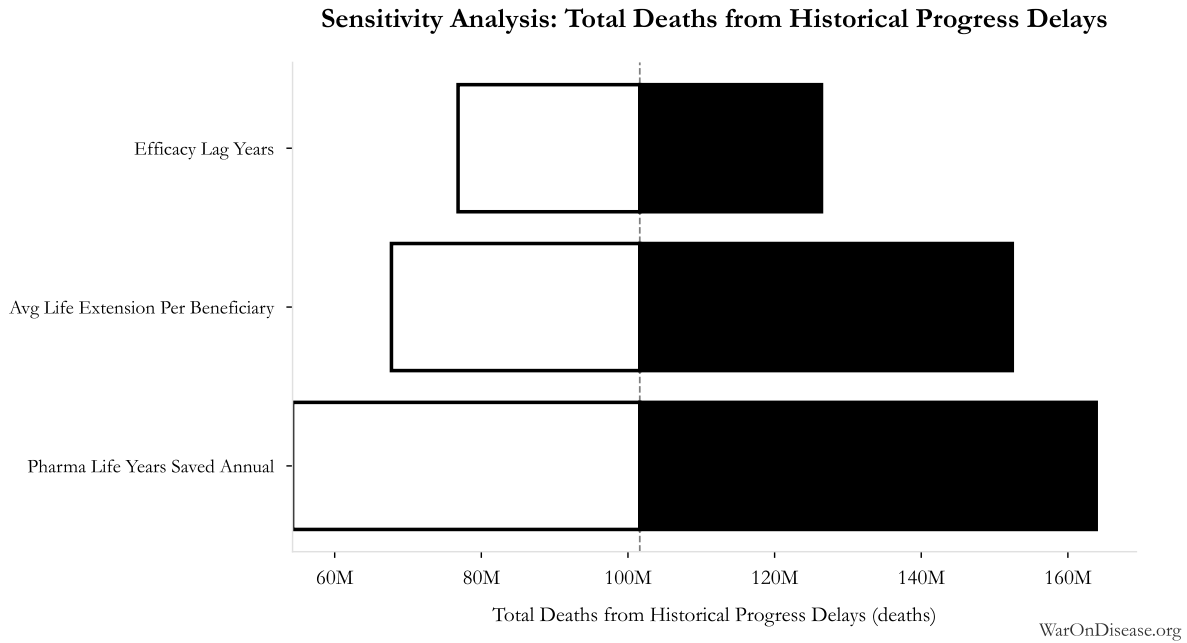


Figure 3: Monte Carlo Distribution: Total Deaths from Historical Progress Delays (10,000 simulations)

Simulation Results Summary: Total Deaths from Historical Progress Delays

Statistic	Value
Baseline (deterministic)	102M
Mean (expected value)	107M
Median (50th percentile)	97.3M
Standard Deviation	53.0M
90% Confidence Interval	[36.9M, 214M]

The histogram shows the distribution of Total Deaths from Historical Progress Delays across 10,000 Monte Carlo simulations. The CDF (right) shows the probability of the outcome exceeding any given value, which is useful for risk assessment.



Combined, these represent **7.94B DALYs (95% CI: 4.43B DALYs-12.1B DALYs) Disability-Adjusted Life Years (DALYs)** when adjusted for morbidity. All estimates include Monte Carlo confidence intervals.

Valuing these lost years at a conservative global Value of a Statistical Life Year (VSLY), we find a cumulative economic deadweight loss of approximately **\$1.19 quadrillion (95% CI: \$443T-\$2.41 quadrillion)** (2024 USD). The study concludes that the societal cost of **Type II Regulatory Errors** (delayed access to effective therapies) exceeds the averted cost of **Type I Regulatory Errors** (market access for ineffective therapies) by a factor of **3.07k:1 (95% CI: 2.88k:1-3.12k:1)**.

2 Scale

9/11: 2.98k people dead. We spent \$8 trillion in response.

Holocaust: 6 million dead.

Efficacy lag: 102M deaths (95% CI: 36.9M deaths-214M deaths) dead. That’s 34.1k 9/11s (95% CI: 12.4k 9/11s-71.8k 9/11s), or 17 Holocausts.

We paid \$4.84T (95% CI: \$3.42T-\$6.62T) (lower bound - Phase 2/3 costs only) to cause 34.1k 9/11s (95% CI: 12.4k 9/11s-71.8k 9/11s).

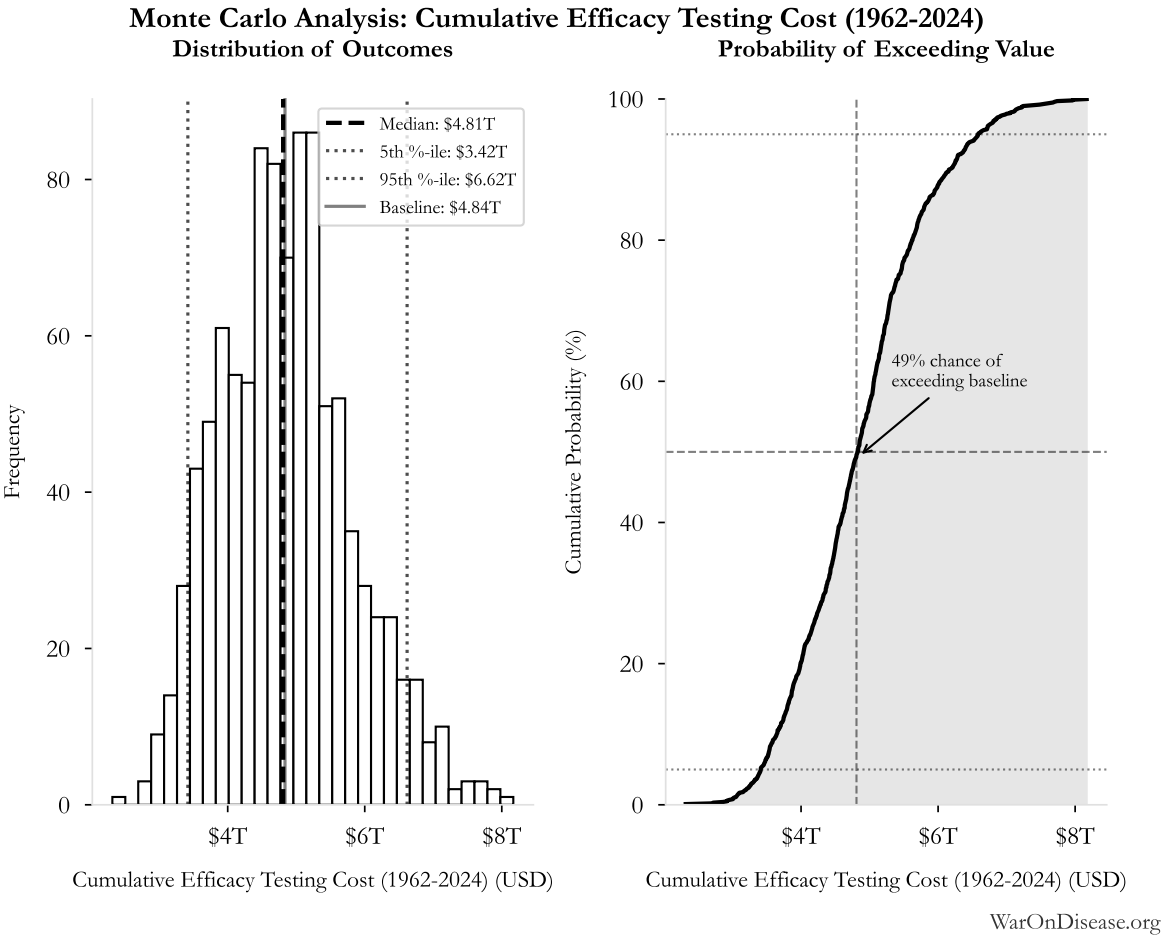
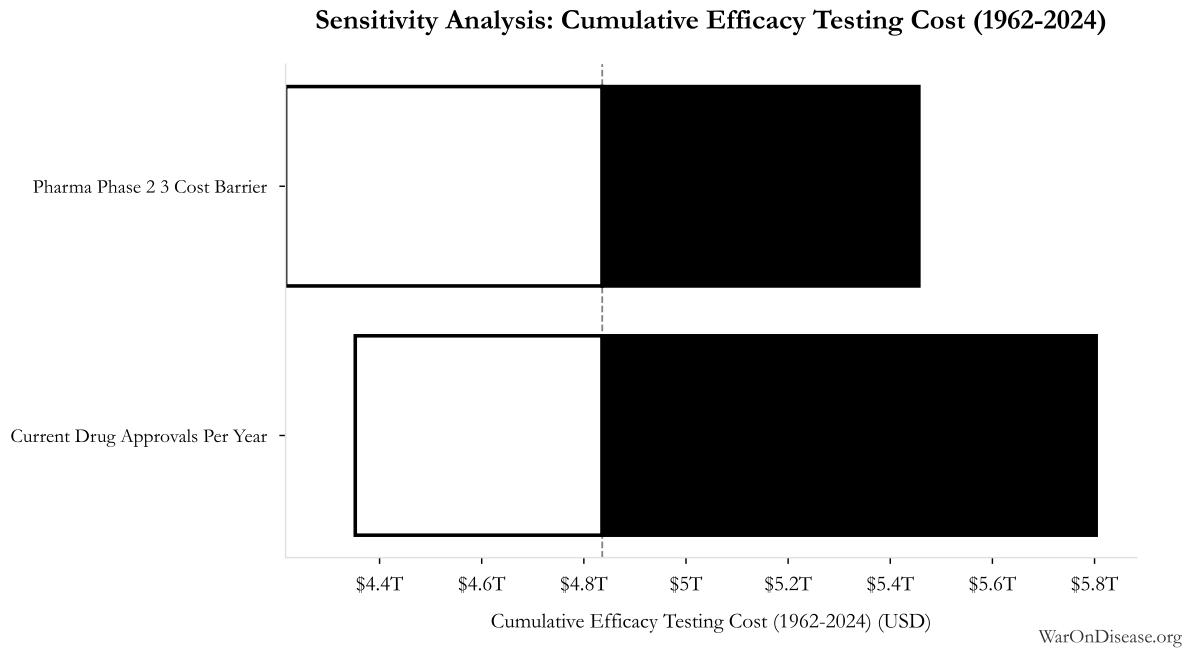


Figure 4: Monte Carlo Distribution: Cumulative Efficacy Testing Cost (1962-2024) (10,000 simulations)

Simulation Results Summary: Cumulative Efficacy Testing Cost (1962-2024)

Statistic	Value
Baseline (deterministic)	\$4.84T
Mean (expected value)	\$4.88T
Median (50th percentile)	\$4.81T
Standard Deviation	\$977B
90% Confidence Interval	[\$3.42T, \$6.62T]

The histogram shows the distribution of Cumulative Efficacy Testing Cost (1962-2024) across 10,000 Monte Carlo simulations. The CDF (right) shows the probability of the outcome exceeding any given value, which is useful for risk assessment.



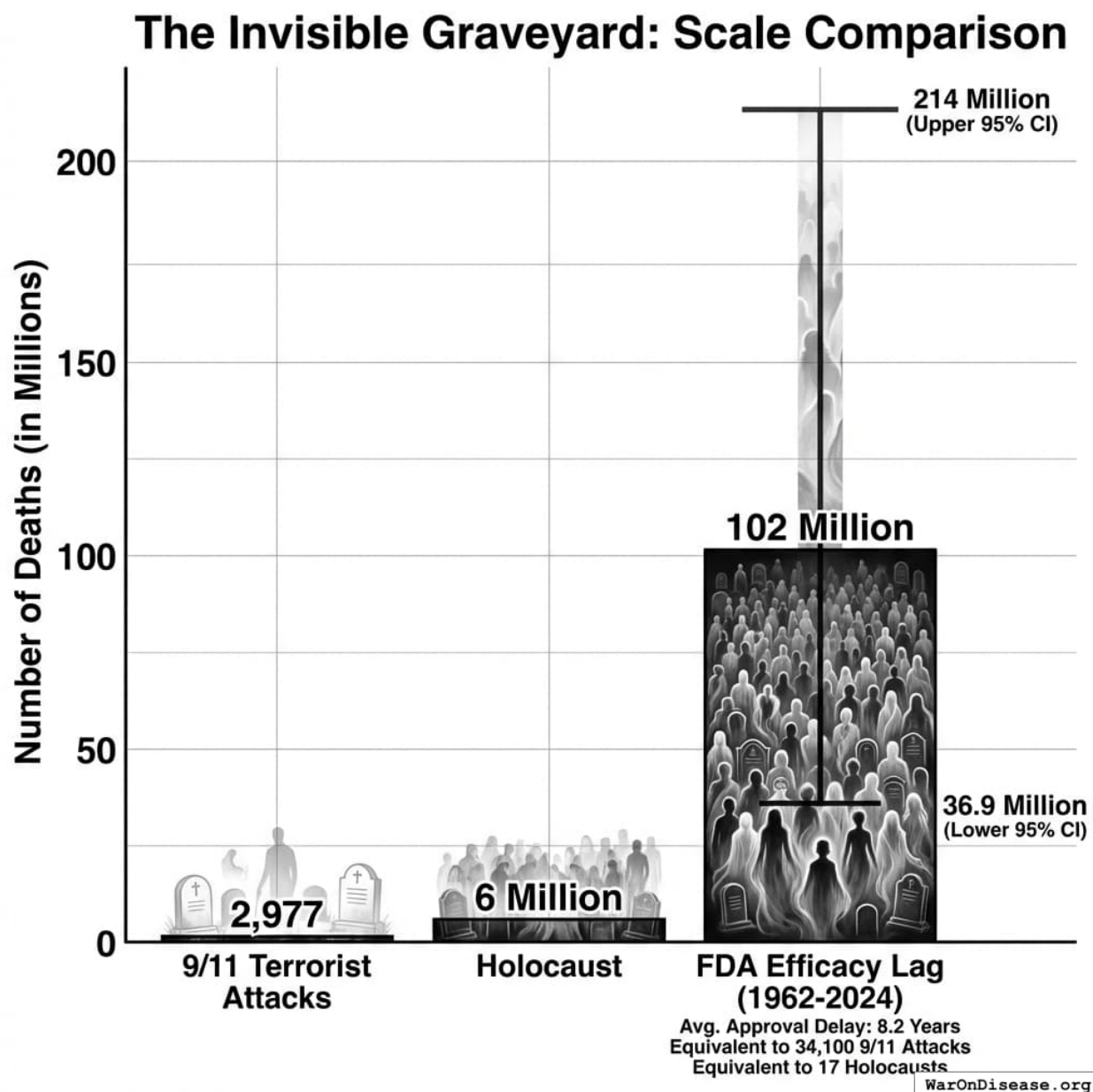


Figure 5: Bar chart with ghostly imagery comparing 9/11 deaths (2,977), Holocaust deaths (6 million), and FDA efficacy lag deaths (102 million, 95% CI: 36.9M-214M) with error bars

That's \$1.56B (95% CI: \$1.23B-\$1.89B) per drug for Phase 2/3 efficacy trials, paid by patients through higher drug prices. Before 1962, the AMA's 144k physicians tracked patient outcomes and JAMA published the results. We replaced that with tiny trials on handpicked patients.

Without mandatory pre-market trials, the market wouldn't be blind. Knowing whether drugs work is one of the highest consumer demands imaginable. Organizations like Consumer Reports, JAMA, and independent research institutes would compete to provide rigorous, large-scale efficacy data - with no pharma conflicts of interest, across real-world populations, with ongoing monitoring instead of a pre-approval snapshot.

These are underestimates. They only count delays to drugs that got developed. The \$2.60B (95% CI: \$1.50B-\$4B) approval cost killed other drugs before they started. We can't count deaths prevented by cures that don't exist.

$$N_{9/11,equiv} = \frac{Deaths_{lag,total}}{N_{9/11}} = \frac{102M}{2,980} = 34,100$$

$$\begin{aligned} & \text{where } Deaths_{lag,total} \\ &= Lives_{saved,annual} \times T_{lag} \\ &= 12.4M \times 8.2 \\ &= 102M \end{aligned}$$

$$\text{where } Lives_{saved,annual} = \frac{LY_{saved,annual}}{T_{ext}} = \frac{149M}{12} = 12.4M$$

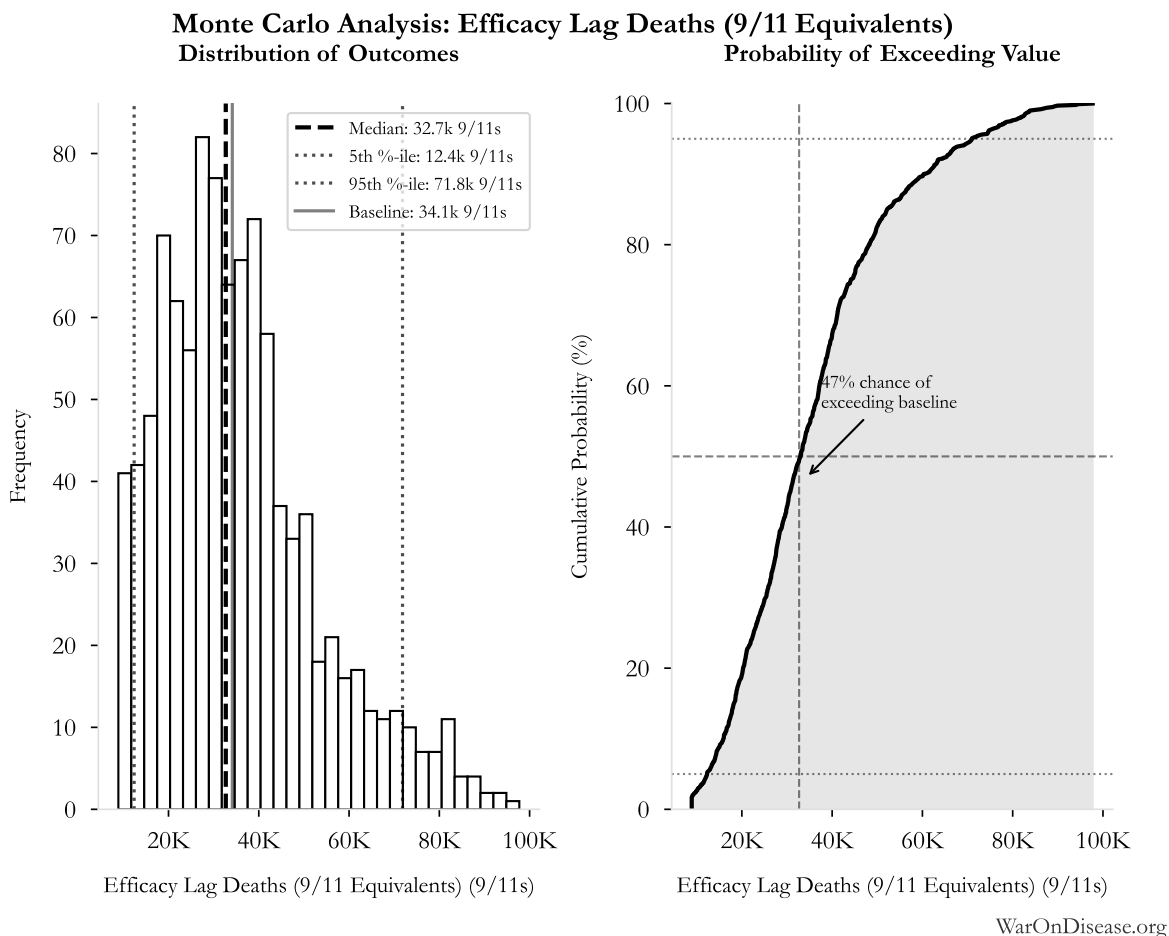
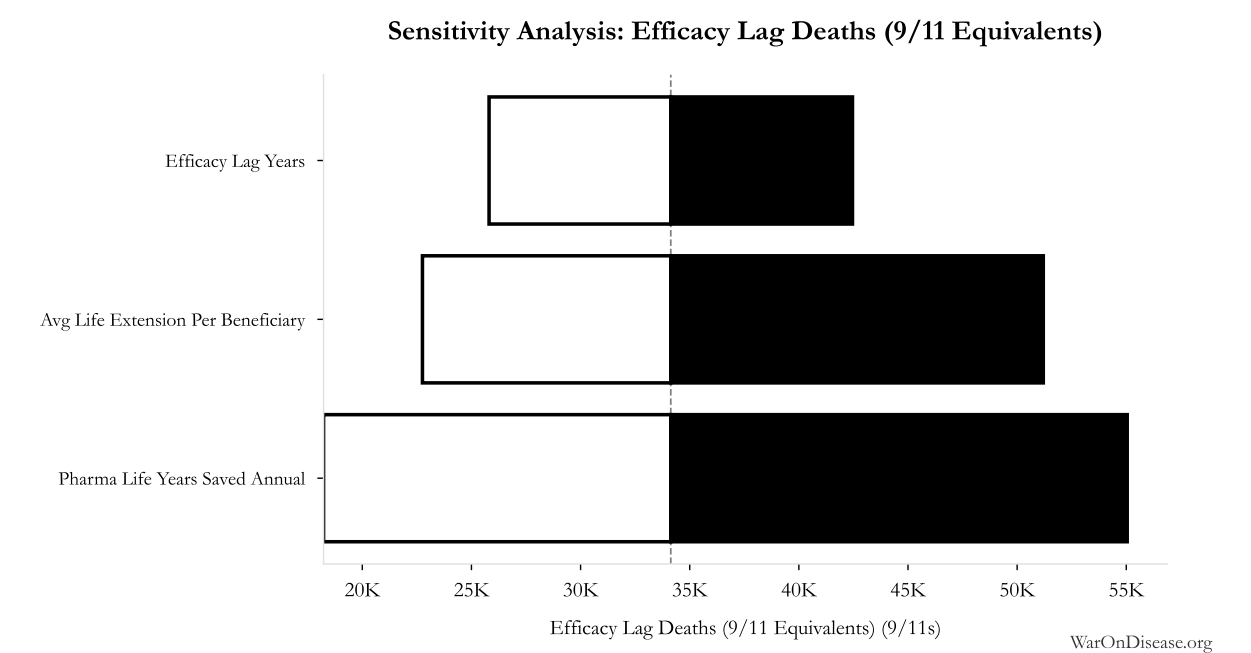


Figure 6: Monte Carlo Distribution: Efficacy Lag Deaths (9/11 Equivalents) (10,000 simulations)

Simulation Results Summary: Efficacy Lag Deaths (9/11 Equivalents)

Statistic	Value
Baseline (deterministic)	34.1k
Mean (expected value)	36.0k
Median (50th percentile)	32.7k
Standard Deviation	17.8k
90% Confidence Interval	[12.4k, 71.8k]

The histogram shows the distribution of Efficacy Lag Deaths (9/11 Equivalents) across 10,000 Monte Carlo simulations. The CDF (right) shows the probability of the outcome exceeding any given value, which is useful for risk assessment.



Sensitivity Indices for Efficacy Lag Deaths (9/11 Equivalents)

Regression-based sensitivity showing which inputs explain the most variance in the output.

Input Parameter	Sensitivity Coefficient	Interpretation
Existing Drugs Efficacy Lag Deaths Total	1.0000	Strong driver

Interpretation: Standardized coefficients show the change in output (in SD units) per 1 SD change in input. Values near ± 1 indicate strong influence; values exceeding ± 1 may occur with correlated inputs.

3 Introduction

The modern pharmaceutical regulatory paradigm relies on a binary licensure model: a drug is either “safe and effective” (approved) or “unsafe/ineffective” (prohibited). While Phase I trials typically establish safety within 2.3 years, the requirement to prove statistical efficacy (Phase II/III) extends the pre-market timeline by an additional **8.2 years (95% CI: 4.85 years-11.5 years)** on average.

This study evaluates the **Bifurcated Regulatory Model (BRM)**, defined as “Safety-First / Efficacy-Later”, to measure the “Invisible Graveyard”: the population that dies during the regulatory latency period between safety verification and final approval.

The Efficacy Lag: Where Patients Die Waiting

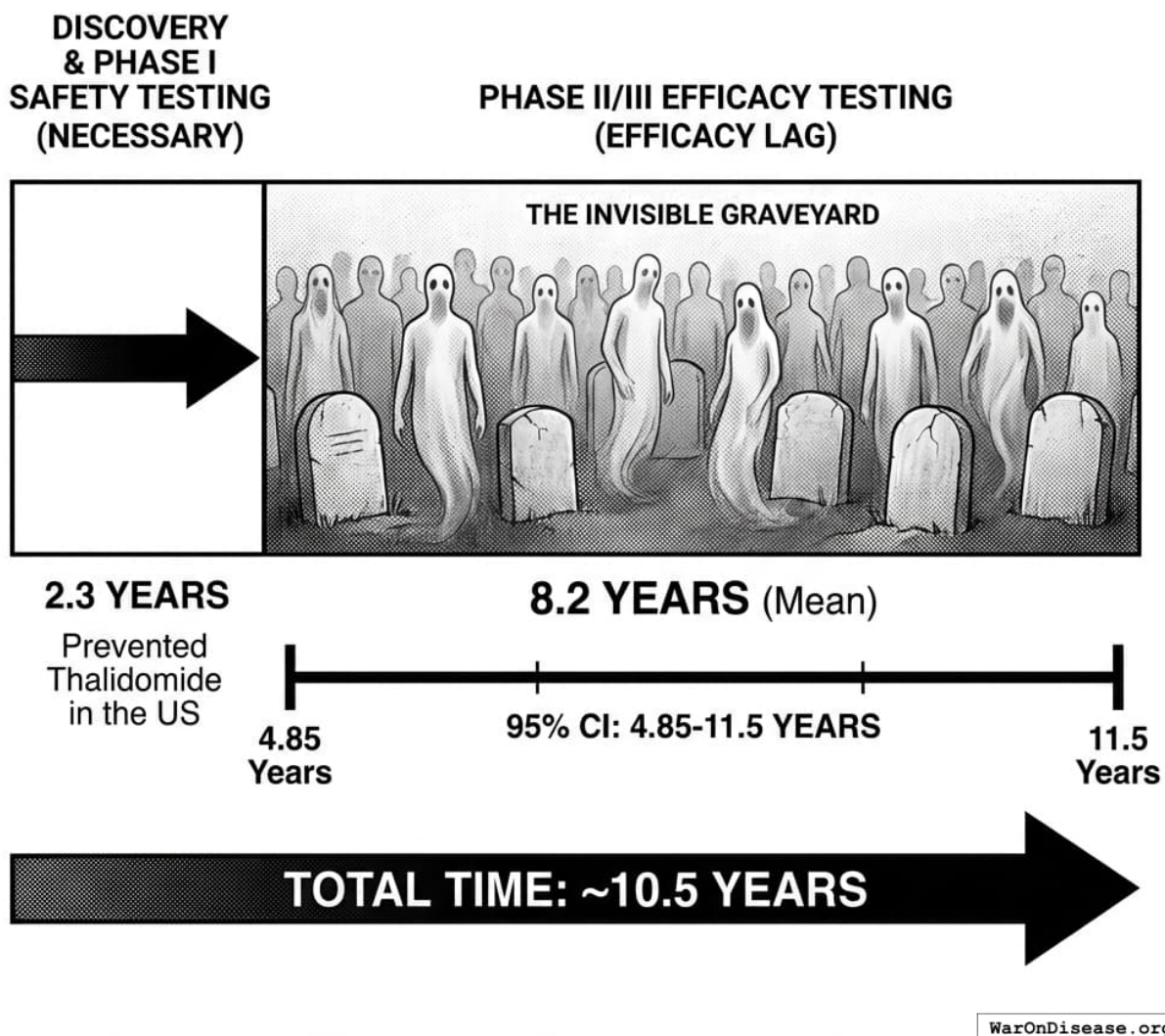


Figure 7: Timeline diagram showing Phase I safety (2.3 years, necessary) vs Phase II/III efficacy lag (8.2 years, 95% CI: 4.85-11.5 years) where patients die waiting

4 Literature Review: The Drug Lag Debate

4.1 Foundational Economic Analysis

The regulatory cost of FDA efficacy requirements was first rigorously quantified by Peltzman¹³⁸, who estimated that the 1962 Kefauver-Harris Amendments reduced the flow of new drugs by 50-60%. His analysis concluded that the costs of reduced pharmaceutical innovation substantially exceeded any benefits from keeping ineffective drugs off the market, resulting in net welfare losses to society.

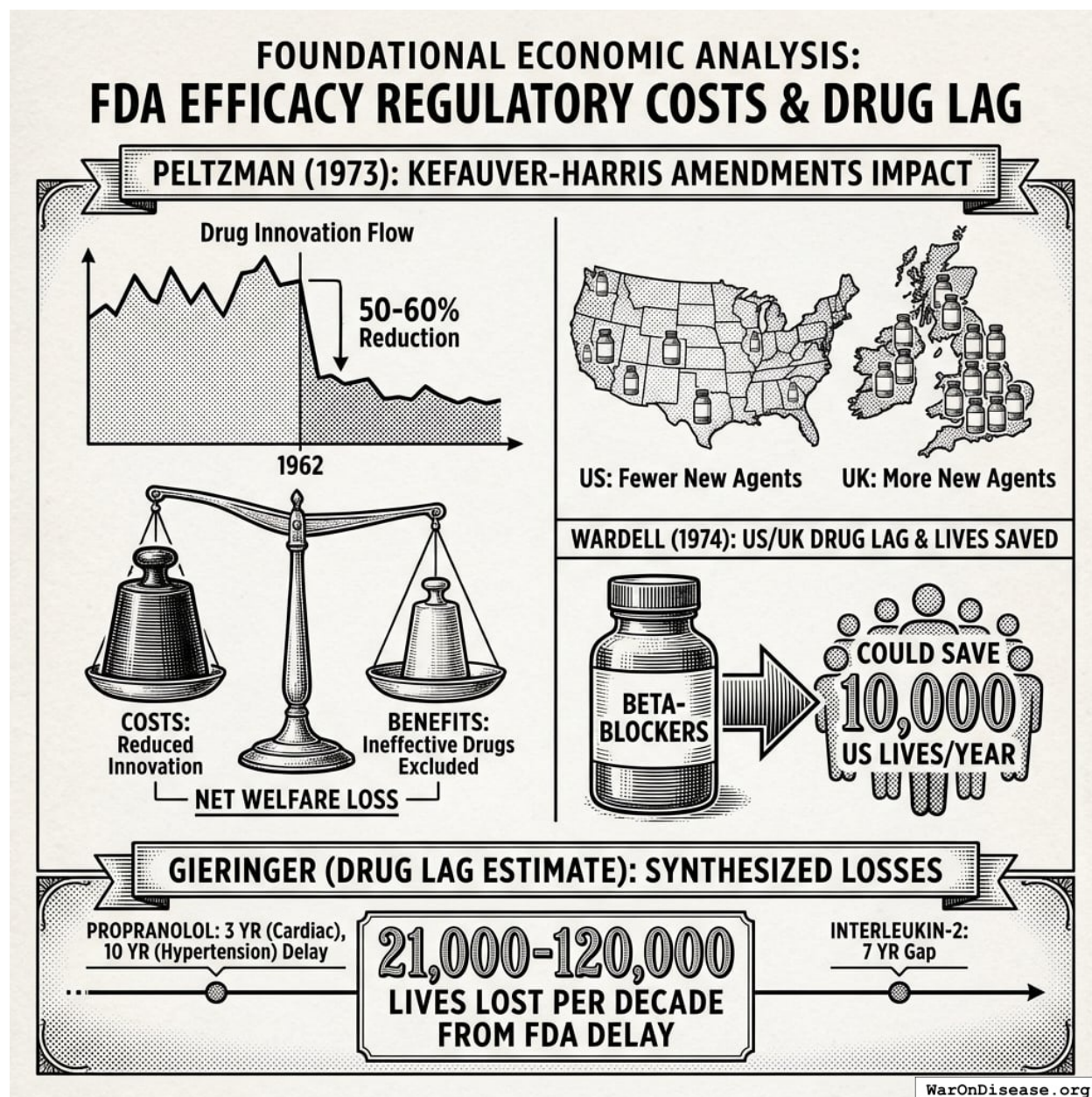


Figure 8: A comparison of regulatory impacts on pharmaceutical innovation, showcasing the estimated 50-60% reduction in new drug flow alongside mortality estimates and specific international approval time lags.

Wardell¹³⁹ documented the emerging “drug lag” between US and UK drug approvals, finding that the UK had access to significantly more new therapeutic agents. His estimate that beta-blockers alone could save 10,000 American lives annually if approved became a landmark finding in regulatory economics.

Gieringer¹⁴⁰ synthesized these estimates, calculating 21,000-120,000 lives lost per decade from FDA delay. His work documented specific drug delays: propranolol (approved in the US 3 years after Europe for cardiac use, 10 years later for hypertension), interleukin-2 (7-year gap), and numerous other therapeutics.

4.2 The Current Debate

Contemporary research continues to find significant regulatory costs. The Tufts Center for the Study of Drug Development documents development timelines of 10.5 years (95% CI: 6 years-12 years) and costs of \$2.60B (95% CI: \$1.50B-\$4B) per approved drug. BIO’s clinical development success rates show only 10% of drugs entering Phase I ultimately reach patients.

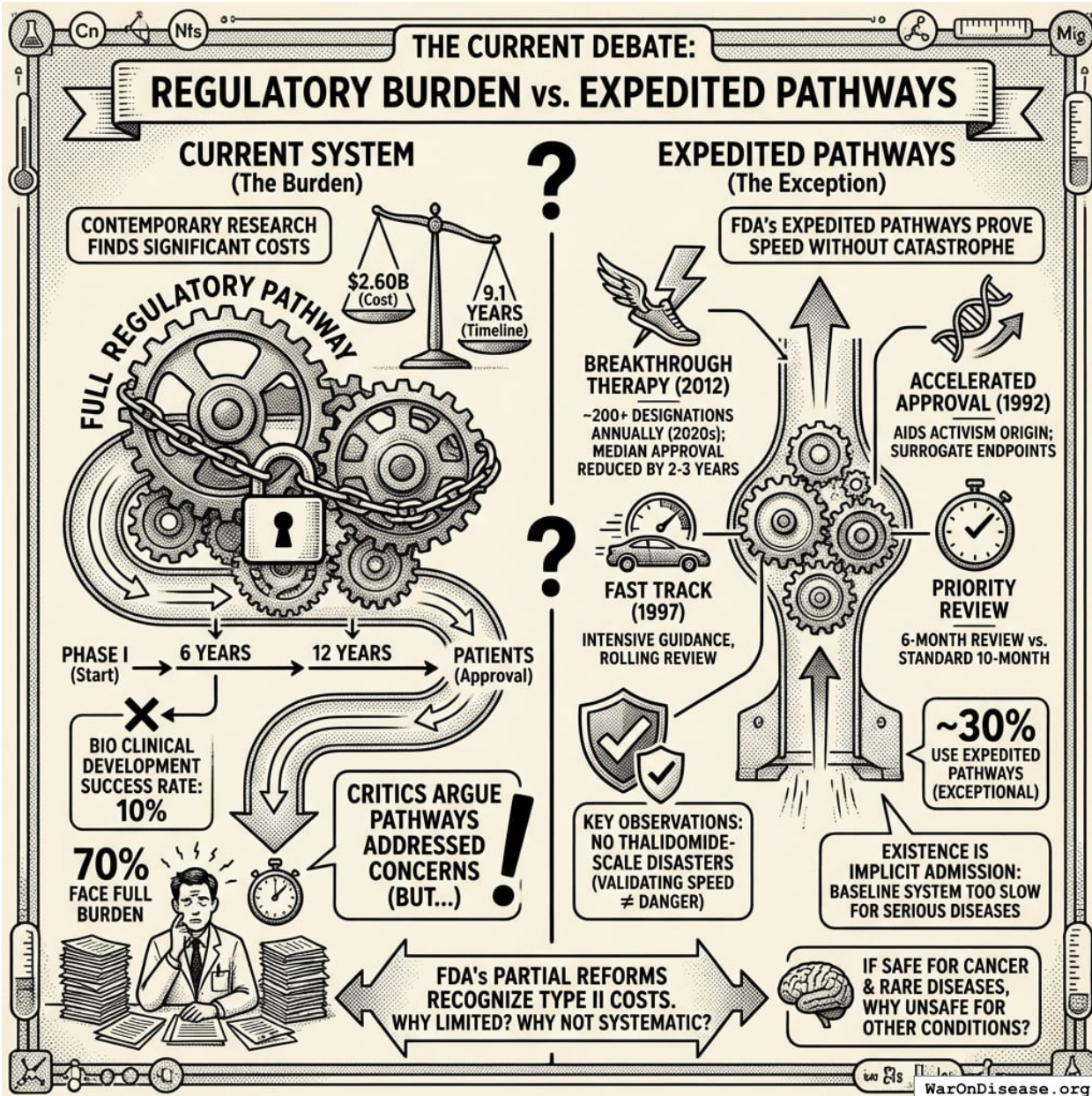


Figure 9: A comparison between the FDA's standard regulatory process and expedited pathways, showing the difference in review timelines and the distribution of drug approvals.

Critics argue that faster approval pathways (breakthrough therapy designation, accelerated approval) have addressed these concerns. However, these pathways actually support our argument:

FDA's Expedited Pathways Prove Speed is Possible Without Catastrophe:

1. **Breakthrough Therapy Designation (2012):** ~200+ designations annually by 2020s, median approval time reduced by 2-3 years for qualifying drugs
2. **Accelerated Approval (1992):** Born from AIDS activism; allows approval based on surrogate endpoints
3. **Fast Track (1997):** Intensive FDA guidance and rolling review

4. **Priority Review:** 6-month review vs. standard 10-month

Key observations: - These pathways have NOT produced Thalidomide-scale disasters, validating that speed = danger - They remain *exceptional* rather than default: ~30% of approvals use expedited pathways; 70% face full regulatory burden - Their existence is an implicit admission that the baseline system is too slow for serious diseases - If expedited pathways are safe for cancer and rare diseases, why are they unsafe for other conditions?

The FDA's partial reforms prove the system recognizes Type II costs exist. The question is why the recognition is limited to a subset of diseases rather than systematically applied.

4.3 **Empirical Case Studies: Demonstrating the Causal Mechanism**

The theoretical claim that regulatory delay causes mortality requires empirical grounding. Three case studies demonstrate the mechanism operates in practice:

1. **Beta-Blockers (1964-1976): The Classic Drug Lag**

Propranolol, the first beta-blocker for treating angina and hypertension, was approved in the UK in 1964. US approval came in 1967 for minor uses, but not until 1973 (angina) and 1976 (hypertension) for cardiovascular indications. Wardell estimated approximately 10,000 Americans died annually during this delay, as the FDA's doors were "essentially closed to cardiovascular drugs for an entire decade"¹³⁹. This single drug's regulatory lag may have caused more American deaths than all other drug-related deaths in that century.

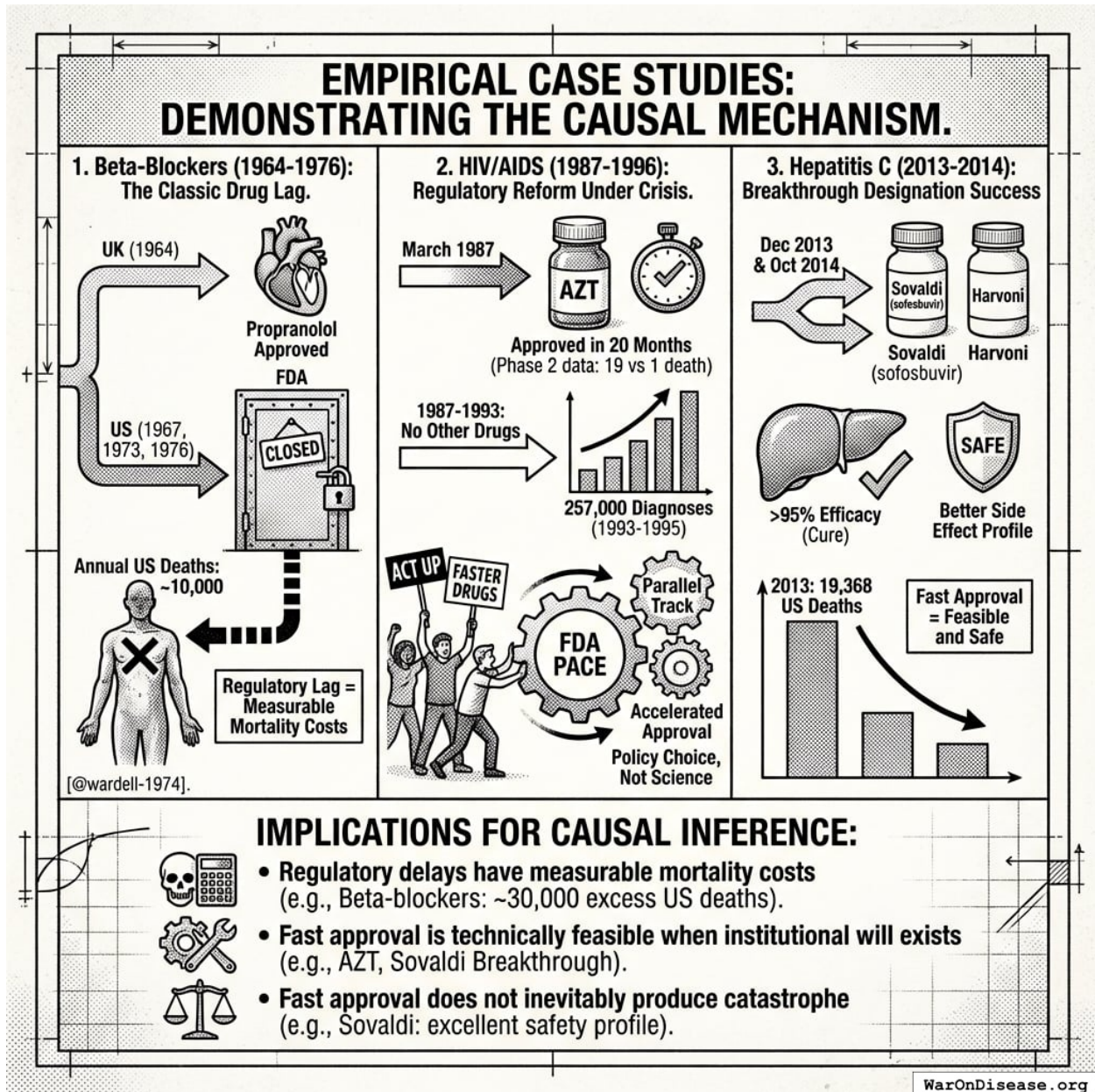


Figure 10: A comparative summary of regulatory timelines, mortality statistics, and safety outcomes across three major pharmaceutical case studies: Beta-Blockers, HIV/AIDS, and Hepatitis C.

2. HIV/AIDS (1987-1996): Regulatory Reform Under Crisis

The AIDS epidemic demonstrated that regulatory speed is a policy choice. AZT was approved in March 1987 in a record 20 months, without a Phase 3 trial, after Phase 2 showed 19 placebo deaths vs. 1 treatment death¹⁴¹. This proves expedited approval is technically feasible. However, from 1987-1993, no other AIDS drugs were approved, despite 257,000 diagnoses in 1993-1995 alone. ACT UP activism forced regulatory reforms (Parallel Track, Accelerated Approval), proving that the FDA's pace reflects institutional priorities, not immutable scientific requirements.

3. Hepatitis C (2013-2014): Breakthrough Designation Success

Sovaldi (sofosbuvir) received FDA Breakthrough Therapy designation and was approved December 2013, with Harvoni following in October 2014. These drugs cure HCV in 12 weeks with >95% efficacy. In 2013, HCV caused 19,368 US deaths. Critically, despite rapid approval, no Thalidomide-scale disaster occurred. The drugs’ side effect profile was actually better than prior interferon-based treatments. This demonstrates that fast approval of transformative drugs is both possible and safe.

Implications for Causal Inference:

These cases establish that: - Regulatory delays have measurable mortality costs (beta-blockers: if Wardell’s 10,000/year estimate holds, the 3-year US delay implies ~30,000 excess deaths) - Fast approval is technically feasible when institutional will exists (AZT: 20 months; Sovaldi: Breakthrough pathway) - Fast approval does not inevitably produce catastrophe (Sovaldi: excellent safety profile)

The counterfactual is not purely speculative: we observe the mechanism operating in discrete cases where data is available.

5 Methodology & Data

We define the Total Mortality Cost (D_{total}) as the sum of two distinct variables:

$$D_{total} = D_{lag} + D_{void}$$

5.1 Variable Definitions

- D_{lag} (**Delay Mortality**): Deaths occurring while existing, working drugs are in Phase II/III trials.
- D_{void} (**Innovation Loss**): Deaths occurring because high regulatory costs prevented the development of potential cures (The “Innovation Tax”).

5.2 Theoretical Upper Bound: What’s Eventually Preventable?

Before calculating regulatory delay costs, we must establish what percentage of deaths are *theoretically* preventable with sufficient biomedical advancement. This sets the upper bound for any intervention.

! Methodological Note: Distinguishing Current vs. Theoretical Preventability

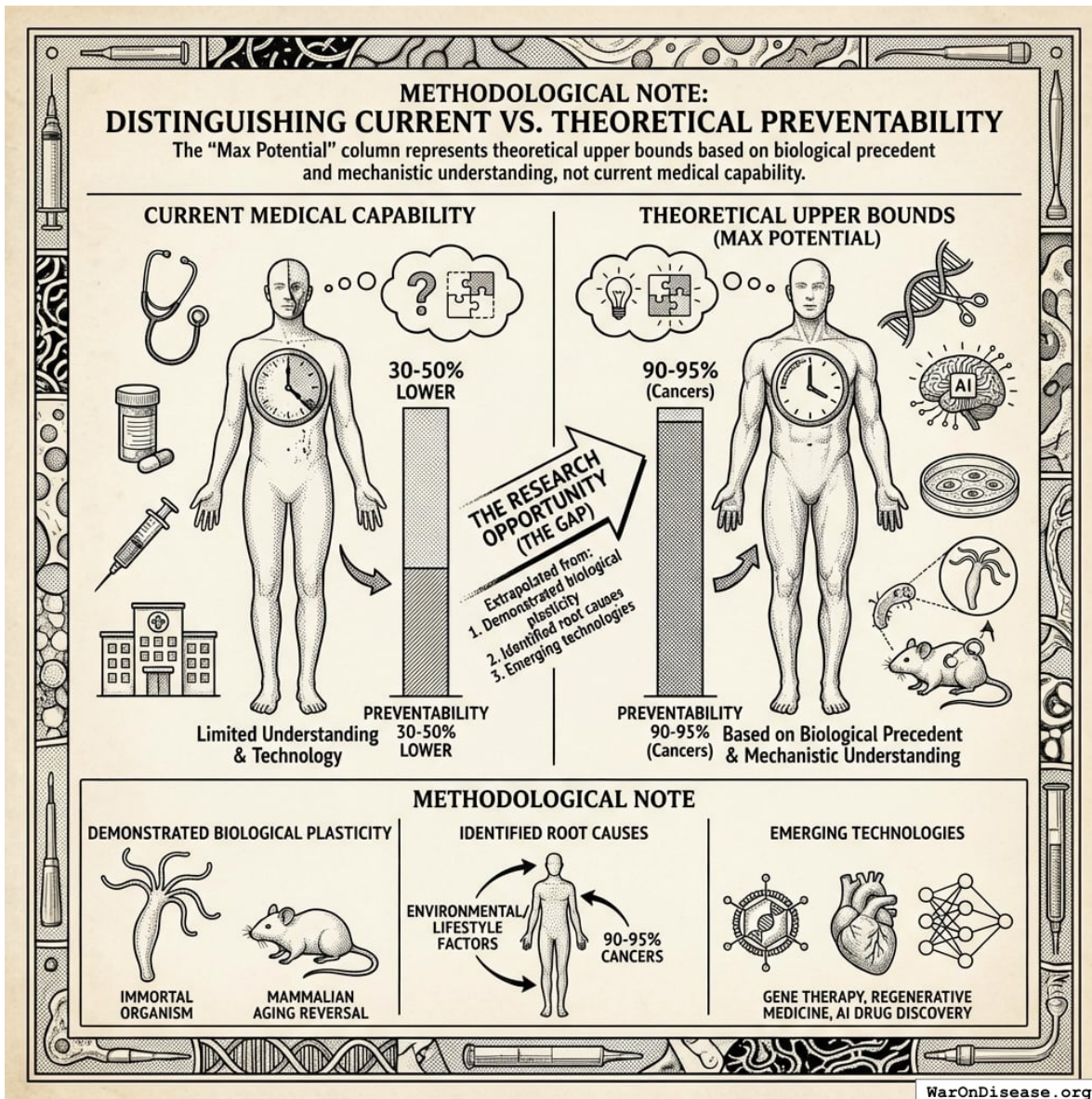


Figure 11: A comparison bar chart showing the gap between current medical preventability and the theoretical maximum potential, highlighting the 30-50% research opportunity margin.

The "Max Potential" column represents **theoretical upper bounds** based on biological precedent and mechanistic understanding, not current medical capability. These estimates extrapolate from:

1. **Demonstrated biological plasticity** (organisms that don't age, mammalian aging reversal)
2. **Identified root causes** (90-95% of cancers have environmental/lifestyle roots)
3. **Emerging technologies** (gene therapy, regenerative medicine, AI drug discovery)

Current preventability is typically 30-50% lower than theoretical maximum. The gap represents the research opportunity.

5.2.1 Disease Burden by Category

Using WHO Global Burden of Disease¹⁴² data, we categorize annual deaths:

Category	% of Deaths	Current	Max Potential	Source for Max Estimate
Cardiovascular	26.0%	50%	95%	WHO: 80-90% preventable ¹⁴³
Cancer	18.9%	69%	95%	90-95% environmental/lifestyle roots ¹⁴⁴
Aging-related	23.2%	5%	99%	Mammalian aging reversal demonstrated ¹⁴⁵
Accidents	8.0%	30%	60%	WHO: largely preventable ¹⁴⁶
Metabolic	6.3%	70%	98%	Diabetes reversal via gene therapy ¹⁴⁷
Respiratory	4.3%	60%	90%	WHO: 80% of COPD preventable ²
Neurodegenerative	3.6%	10%	80%	Stem cell therapy potential ¹⁴⁸
Infectious	1.9%	95%	99%	Vaccines + antimicrobials ¹⁴⁹
Other	7.7%	50%	95%	Weighted average of above categories ³

Result: 92.6% (95% CI: 50%-98%) of deaths are eventually avoidable with sufficient research.

5.2.2 Why This Upper Bound? The Biological and Epidemiological Evidence

The “max potential” estimates above are grounded in peer-reviewed research:

1. **Aging has been reversed in mammals.** Yamanaka factor therapy extended remaining lifespan by 109% in aged mice¹⁴⁵ and reversed epigenetic age in human skin cells by 30 years. The mechanisms are understood; we lack only the engineering to apply them safely in humans.

¹Furuyama et al. (2019) used AAV gene therapy to reprogram alpha cells into insulin-producing beta cells, reversing autoimmune diabetes in mice. Max potential extrapolates from root cause addressability.

²WHO estimates 80% of COPD cases preventable through tobacco control and air quality improvements (see WHO COPD Fact Sheet). The 90% max potential conservatively assumes emerging regenerative medicine may address some remaining cases.

³Calculated as weighted average of “Max Potential” estimates for categories with similar biological mechanisms.

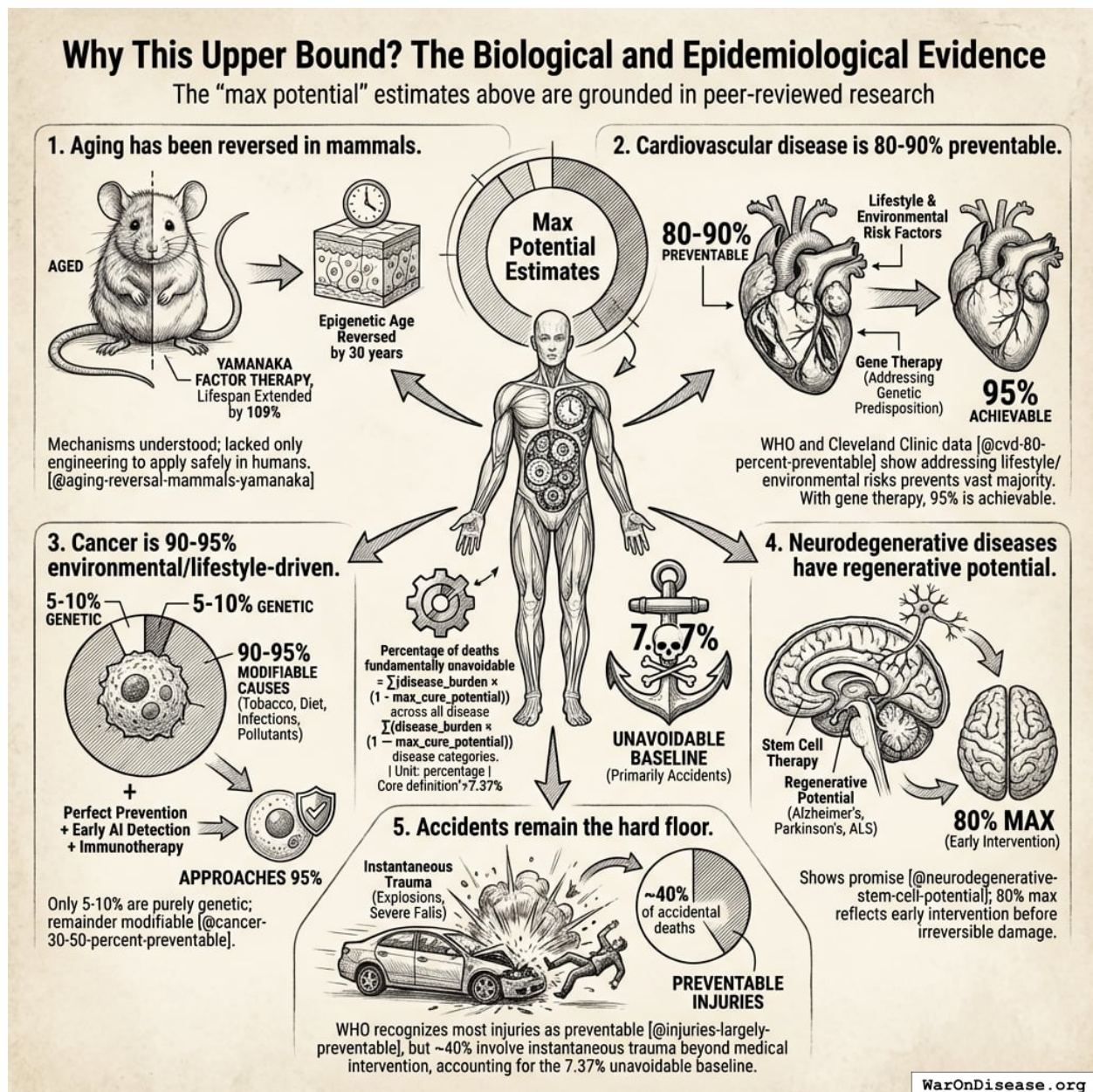


Figure 12: A comparison of the potential prevention and reversal rates across major health categories, including aging, cardiovascular disease, cancer, and neurodegenerative conditions, contrasted against the unavoidable baseline of accidental trauma.

- Cardiovascular disease is 80-90% preventable.** WHO and Cleveland Clinic data¹⁴³ show that addressing lifestyle and environmental risk factors prevents the vast majority of heart attacks and strokes. With gene therapy addressing genetic predisposition, 95% is achievable.
- Cancer is 90-95% environmental/lifestyle-driven.** Only 5-10% of cancers are purely genetic¹⁴⁴; the remainder have modifiable causes (tobacco, diet, infections, pollutants). Perfect prevention + early AI detection + immunotherapy approaches 95%.
- Neurodegenerative diseases have regenerative potential.** Stem cell therapy shows

promise¹⁴⁸ for Alzheimer's, Parkinson's, and ALS. The 80% max reflects early intervention before irreversible damage.

5. **Accidents remain the hard floor.** WHO recognizes most injuries as preventable¹⁴⁶, but ~40% of accidental deaths involve instantaneous trauma (explosions, severe falls) beyond any medical intervention. This accounts for the 7.37% unavoidable baseline.

5.2.3 The 7.37% Floor

The remaining deaths are **fundamentally unavoidable** even with perfect biotechnology:

- Instantaneous traumatic death (e.g., explosions, severe falls)
- Drowning beyond rescue window
- Violence/homicide
- Certain catastrophic accidents

These represent the hard physical limits of medicine. Everything else, including “natural death from old age,” is an engineering problem with engineering solutions.

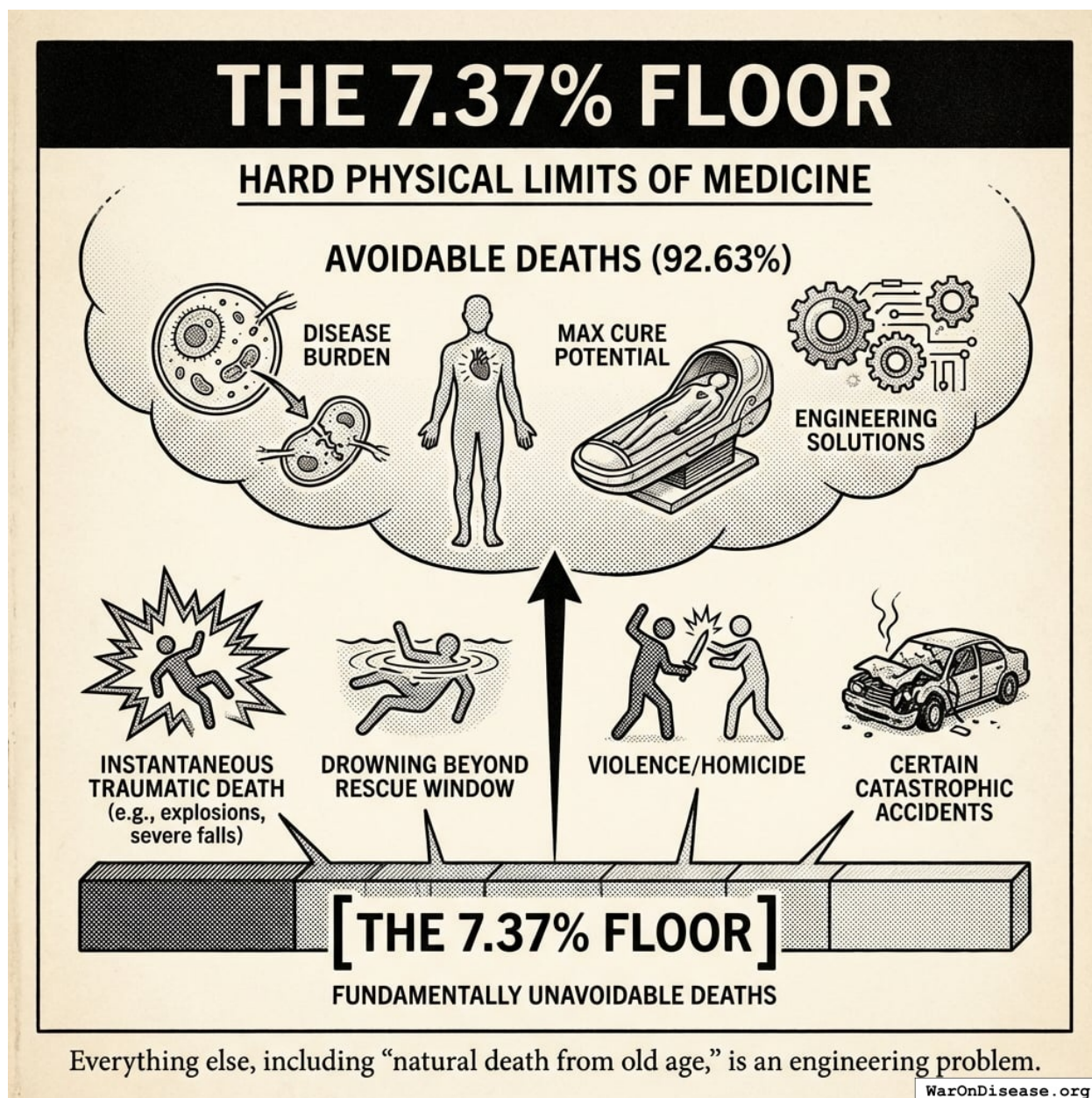


Figure 13: A comparison of mortality categories showing the 7.37% physical floor of unavoidable deaths versus the remaining addressable mortality through engineering.

5.3 Data Sources & Parameterization

1. Development Timelines: *Biotechnology Innovation Organization (BIO) Clinical Development Success Rates 2011–2020.*

- *Verified Metric:* Phase I duration = 2.3 years. Total Time to Market = 10.5 years (95% CI: 6 years-12 years). **Lag = 8.2 years (95% CI: 4.85 years-11.5 years) (95% CI: 4.3-12.1 years)** - wide variance by therapeutic area (oncology ~9y, vaccines ~7y, rare disease ~12+y).
- *Source:* [BIO.org Clinical Development Report](https://www.bioniversitycapital.com/bioorg-clinical-development-report)

2. **Pharmaceutical Impact (Life-Years Saved):** Primary source: Lichtenberg (2019)⁸¹.
- **Primary metric:** 149M life-years (95% CI: 79.4M life-years-240M life-years) saved annually by post-1981 drugs (22 countries, 66 diseases)
 - **Methodology:** 3-way fixed-effects regression (disease-country-year) controlling for confounders
 - **Derived lives saved:** 12.4M deaths (95% CI: 7.60M deaths-18.6M deaths) (assuming 12 years (95% CI: 8 years-18 years) average life extension per beneficiary)

i Life-Years vs. Lives

Lichtenberg measured **life-years saved**, not lives. Converting to “lives” requires assuming average life extension per beneficiary (12 years (95% CI: 8 years-18 years)). Life-years is the more rigorous metric; lives is used for intuitive communication. The uncertainty in the conversion is reflected in the confidence intervals.

Supporting evidence (approximate, for context):

- **Vaccines:** ~4.5M lives/year (WHO estimates 154M lives saved over 50 years)¹⁴⁹
 - **Cardiovascular:** ~3.3M lives/year (Resolve to Save Lives / GBD Data)
 - **Oncology:** ~1.5M lives/year (NBER longevity studies)
3. **Economic Valuation:** *Standard QALY Valuation.*
- **VSLY (Value of a Statistical Life Year):** Standardized at **\$150K (95% CI: \$100K-\$199K)** (consistent with project-wide QALY valuations).

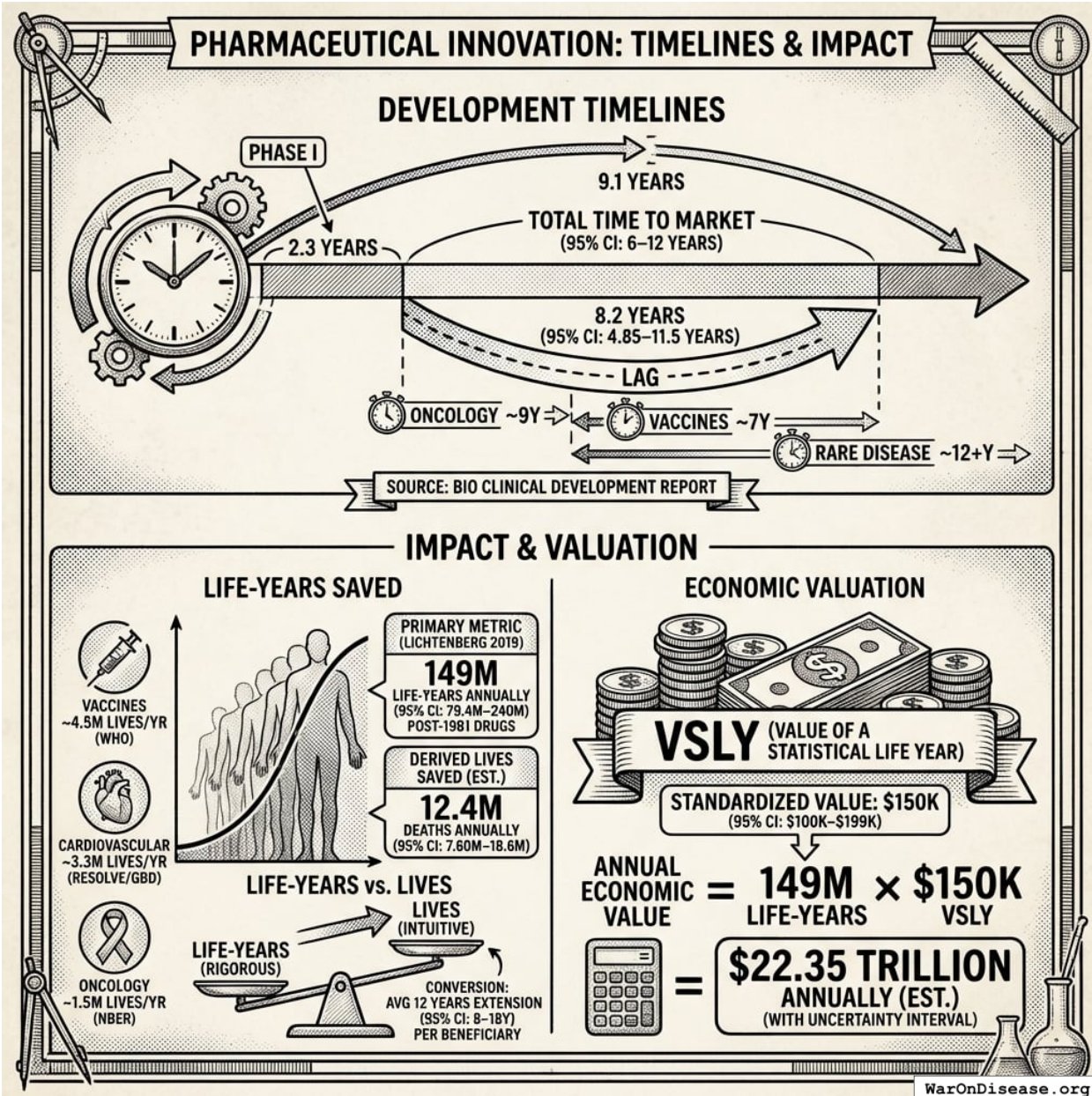


Figure 14: A visualization of the drug development timeline alongside the conversion logic from life-years saved to annual lives saved across major therapeutic areas.

5.4 Uncertainty Quantification Methodology

This analysis employs **Probabilistic Sensitivity Analysis (PSA)** via Monte Carlo simulation to propagate parameter uncertainty through all calculations.

Distribution Selection:

- **Normal:** Symmetric uncertainty around point estimates (e.g., trial duration)
- **Lognormal:** Right-skewed, strictly positive values (costs, relative risks)
- **Beta:** Bounded probabilities [0,1] (success rates, adoption rates)

- **Triangular:** When only min/mode/max available from literature

Propagation Method:

1. Sample N=10,000 draws from each input parameter's distribution
2. Recompute all derived parameters for each Monte Carlo draw
3. Report median and 95% credible intervals (2.5th-97.5th percentiles)

Sensitivity Analysis:

Tornado charts identify which input parameters drive outcome uncertainty by varying each parameter ± 1 standard deviation while holding others at baseline. Standardized regression coefficients (*) enable comparison across parameters with different units.

See [Parameters & Calculations Appendix](#) for complete parameter distributions, formulas, and sensitivity analyses for each calculated value.

6 Results: The Mortality Burden

6.1 Primary Estimate

Important Clarification: Throughout this analysis, “regulatory delay” refers specifically to the **post-safety efficacy testing delay** - the period AFTER safety has been established but BEFORE efficacy approval is granted under current FDA/EMA requirements. This is distinct from safety testing (Phase I), which we consider necessary and effective (as demonstrated by the thalidomide case where safety testing prevented thousands of U.S. deaths).

Methodological Caveat: Cascade Assumption

The primary estimate assumes that the 8.2 years (95% CI: 4.85 years-11.5 years) regulatory delay cascades fully through the biomedical research timeline - i.e., that delaying Drug A by 8.2 years (95% CI: 4.85 years-11.5 years) also delays all downstream research that builds on Drug A's findings by approximately the same amount. This “full cascade” assumption represents a **theoretical upper bound**. In practice, parallel research tracks, international approvals, and adaptive innovation may partially mitigate cascade effects.

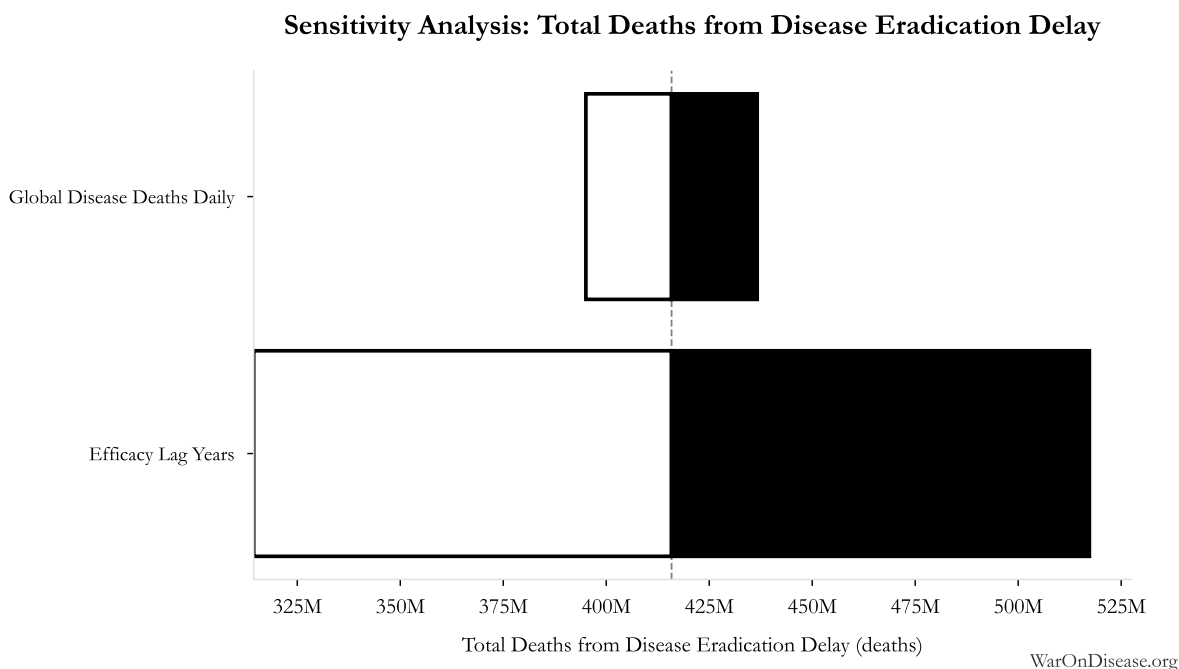
The assumption is not empirically validated at the aggregate level, though individual case studies (beta-blockers, HIV/AIDS, Hepatitis C) demonstrate the mechanism operates in specific instances. The Type II/Type I ratio remains robust even under substantially reduced cascade assumptions (see [sensitivity analysis](#) showing the conclusion holds at 10% regulatory attribution).

Metric	Estimate	Methodology
Total Deaths	416M deaths (95% CI: 225M deaths-630M deaths)	Regulatory delay shifts disease eradication timeline by 8.2 years (95% CI: 4.85 years-11.5 years). Uses WHO global disease mortality rate (150k deaths/day/day).

Finding: The disease eradication delay model estimates **416M deaths (95% CI: 225M deaths-630M deaths) total eventually avoidable deaths**, with **150k deaths/day per day** - greater than the combined casualties of World War I and World War II over the 62-year period.

$$\begin{aligned}
& Deaths_{lag} \\
&= T_{lag} \times Deaths_{disease,daily} \times 338 \\
&= 8.2 \times 150,000 \times 338 \\
&= 416M
\end{aligned}$$

6.1.1 Sensitivity Analysis



Sensitivity Indices for Total Deaths from Disease Eradication Delay

Regression-based sensitivity showing which inputs explain the most variance in the output.

Input Parameter	Sensitivity Coefficient	Interpretation
Efficacy Lag Years	1.1404	Strong driver
Global Disease Deaths Daily	-0.1422	Weak driver

Interpretation: Standardized coefficients show the change in output (in SD units) per 1 SD change in input. Values near ± 1 indicate strong influence; values exceeding ± 1 may occur with correlated inputs.

7 Morbidity Analysis: DALYs and QALYs

Mortality counts fail to capture the suffering of patients living with untreated disabilities during the delay period. We calculated **Disability-Adjusted Life Years (DALYs)** using the formula $DALY = YLL + YLD$.

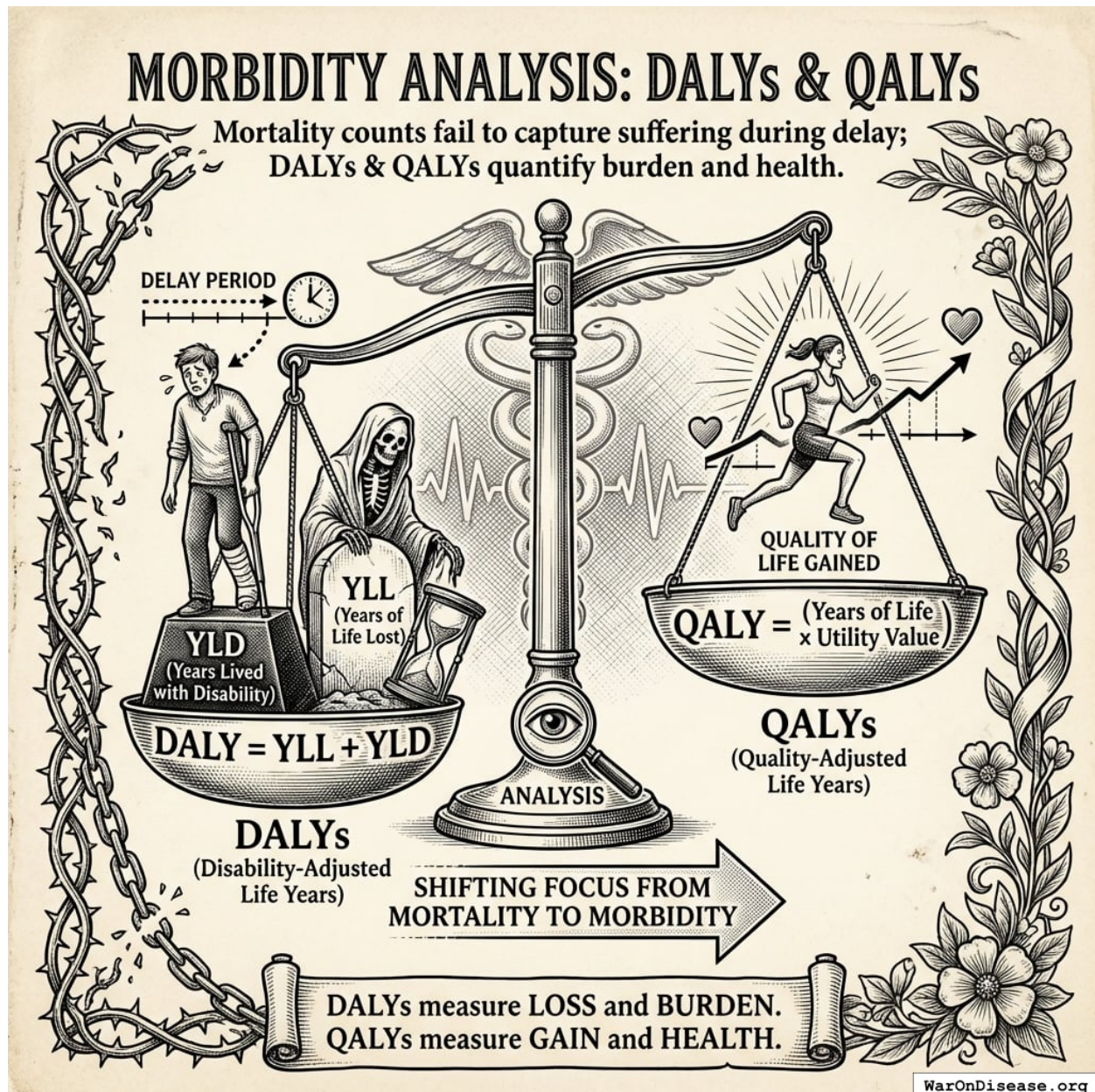


Figure 15: A diagram illustrating how Disability-Adjusted Life Years (DALYs) are calculated by summing Years of Life Lost (YLL) due to premature mortality and Years Lived with Disability (YLD).

7.1 Years of Life Lost (YLL)

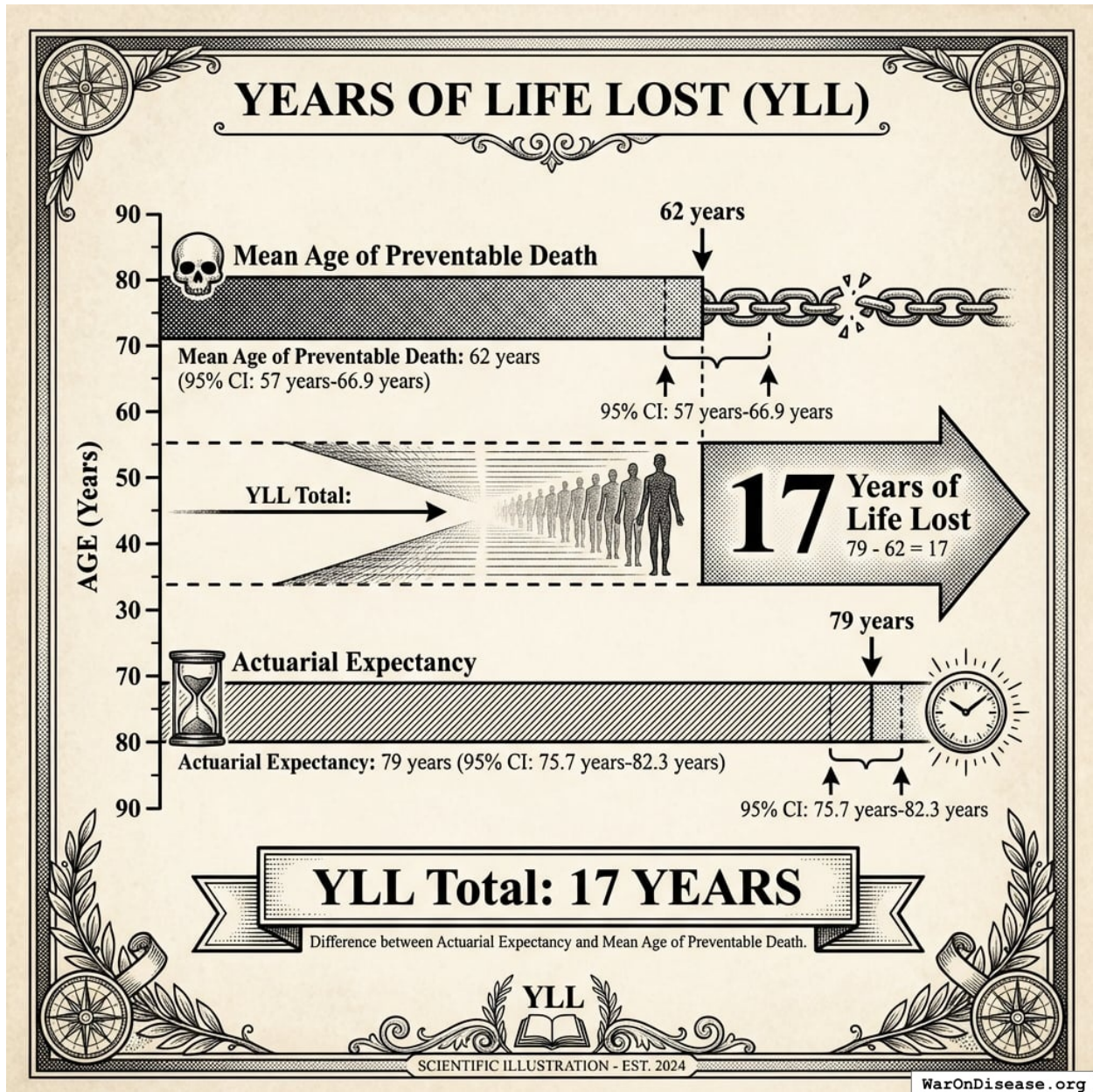


Figure 16: A timeline comparison showing the gap between the mean age of preventable death at 62 years and the actuarial life expectancy of 79 years, representing the Years of Life Lost (YLL).

- **Mean Age of Preventable Death:** 62 years (95% CI: 57 years-66.9 years)
- **Actuarial Expectancy:** 79 years (95% CI: 75.7 years-82.3 years)
- **YLL Total:**

$$\begin{aligned}
& YLL_{lag} \\
&= Deaths_{lag} \times (LE_{global} - Age_{death, delay}) \\
&= 416M \times (79 - 62) \\
&= 7.07B
\end{aligned}$$

$$\begin{aligned}
& \text{where } Deaths_{lag} \\
&= T_{lag} \times Deaths_{disease, daily} \times 338 \\
&= 8.2 \times 150,000 \times 338 \\
&= 416M
\end{aligned}$$

7.2 Years Lived with Disability (YLD)

- **Disability Weight (DW):** 0.35 weight (95% CI: 0.233 weight-0.465 weight) (Weighted average for untreated chronic conditions)
- **Pre-Death Suffering Period:** 6 years (95% CI: 4 years-9 years)
- **YLD Total:**

$$\begin{aligned}
& YLD_{lag} \\
&= Deaths_{lag} \times T_{suffering} \times DW_{chronic} \\
&= 416M \times 6 \times 0.35 \\
&= 873M
\end{aligned}$$

$$\begin{aligned}
& \text{where } Deaths_{lag} \\
&= T_{lag} \times Deaths_{disease, daily} \times 338 \\
&= 8.2 \times 150,000 \times 338 \\
&= 416M
\end{aligned}$$

7.3 Cumulative DALY Burden

$$DALYs_{lag} = YLL_{lag} + YLD_{lag} = 7.07B + 873M = 7.94B$$

$$\text{where } YLL_{lag}$$

$$\begin{aligned} &= Deaths_{lag} \times (LE_{global} - Age_{death, delay}) \\ &= 416M \times (79 - 62) \\ &= 7.07B \end{aligned}$$

$$\text{where } Deaths_{lag}$$

$$\begin{aligned} &= T_{lag} \times Deaths_{disease, daily} \times 338 \\ &= 8.2 \times 150,000 \times 338 \\ &= 416M \end{aligned}$$

$$\text{where } YLD_{lag}$$

$$\begin{aligned} &= Deaths_{lag} \times T_{suffering} \times DW_{chronic} \\ &= 416M \times 6 \times 0.35 \\ &= 873M \end{aligned}$$

$$\text{where } Deaths_{lag}$$

$$\begin{aligned} &= T_{lag} \times Deaths_{disease, daily} \times 338 \\ &= 8.2 \times 150,000 \times 338 \\ &= 416M \end{aligned}$$

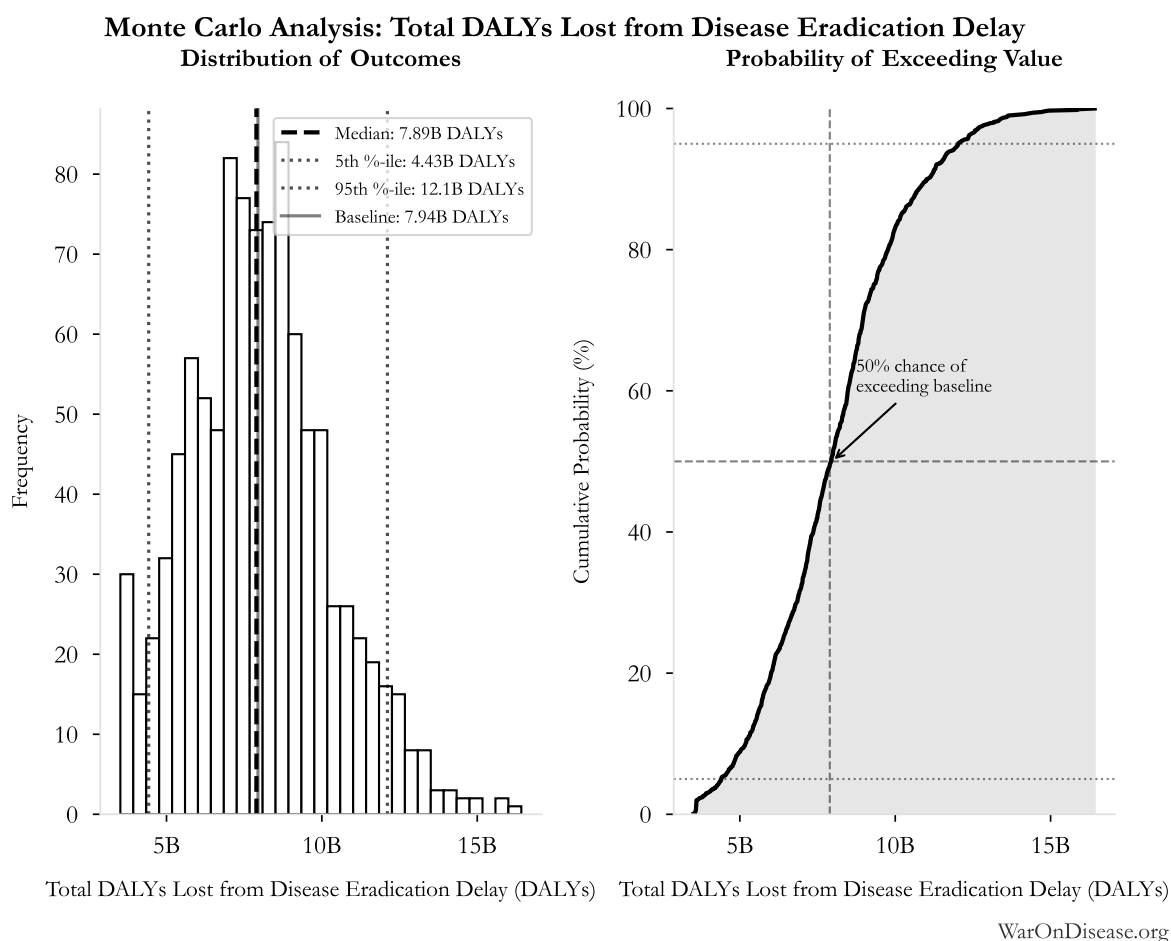


Figure 17: Monte Carlo Distribution: Total DALYs Lost from Disease Eradication Delay (10,000 simulations)

Simulation Results Summary: Total DALYs Lost from Disease Eradication Delay

Statistic	Value
Baseline (deterministic)	7.94B
Mean (expected value)	8.05B
Median (50th percentile)	7.89B
Standard Deviation	2.31B
90% Confidence Interval	[4.43B, 12.1B]

The histogram shows the distribution of Total DALYs Lost from Disease Eradication Delay across 10,000 Monte Carlo simulations. The CDF (right) shows the probability of the outcome exceeding any given value, which is useful for risk assessment.

Interpretation: The regulatory framework has effectively deleted **7.94B DALYs (95% CI: 4.43B DALYs-12.1B DALYs)** billion years of healthy human life.

7.4 Years Lived with Disability - Treatment Beneficiaries

The YLD calculation above captures suffering *before death* for those who ultimately died from delayed treatments. However, a much larger population - the **982M people (95% CI: 827M people-1.15B people) people annually who receive chronic disease treatment** - also suffered during the 8.2 years (95% CI: 4.85 years-11.5 years) delay before their treatments became available.

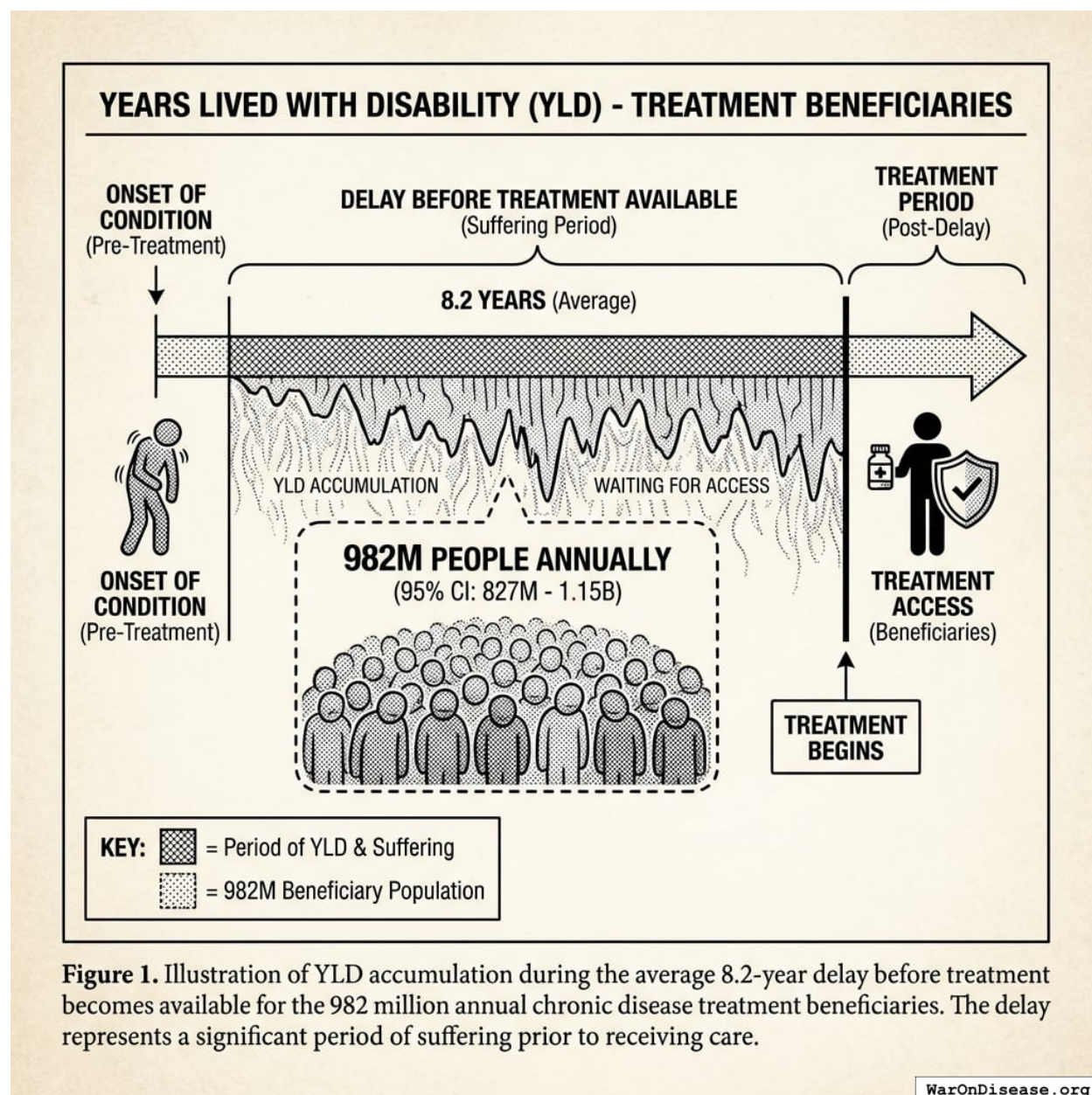


Figure 18: A timeline visualizing the 8.2-year delay period experienced by 982 million people before receiving treatment, illustrating the accumulation of years lived with disability (YLD).

i Distinction: Mortality vs. Morbidity Burden

The “12.4M deaths (95% CI: 7.60M deaths-18.6M deaths) lives saved annually” from Lichtenberg’s analysis captures mortality - people who would have *died* without post-1962 drugs. But pharmaceutical treatments primarily improve quality of life for people with *non-terminal* chronic conditions: diabetes, hypertension, depression, COPD, arthritis, and cardiovascular disease.

Treatment beneficiaries vastly exceed mortality beneficiaries.

Data source: IQVIA reports that global pharmaceutical use reached 1.8 trillion days of therapy in 2019, with 71% for chronic conditions (diabetes, CVD, respiratory, cancer)⁵⁴. From this, we estimate approximately 982M people (95% CI: 827M people-1.15B people) unique patients receive chronic disease treatment annually.

Treatment beneficiary YLD calculation:

$$\begin{aligned}
 YLD_{treat_delay} &= N_{treated} \times T_{lag} \times \Delta DW_{treat} \\
 &= 982M \times 8.2 \times 0.25 \\
 &= 2.01B \\
 \text{where } N_{treated} &= DOT_{chronic} \times 0.000767 \\
 &= 1.28T \times 0.000767 \\
 &= 982M
 \end{aligned}$$

Interpretation: Each year, patients receiving treatment for chronic conditions would have collectively avoided **2.01B DALYs (95% CI: 661M DALYs-4.41B DALYs) of disability** if those treatments had been available 8.2 years (95% CI: 4.85 years-11.5 years) earlier.

Table 9: Comparison of mortality vs. morbidity burden from regulatory delay

Metric	Annual Burden	Source
Lives saved (mortality)	12.4M deaths (95% CI: 7.60M deaths-18.6M deaths)	Lichtenberg 2019
Treatment beneficiaries (morbidity)	982M people (95% CI: 827M people-1.15B people)	IQVIA 2024
Ratio	~80:1	Morbidity » mortality

The treatment beneficiary population is approximately 80 times larger than the mortality-focused “lives saved” figure, demonstrating that the **morbidity cost of regulatory delay vastly exceeds the mortality cost.**

8 Economic Valuation

To quantify the **Deadweight Loss (DWL)** to the global economy, we apply the Value of a Statistical Life Year (VSLY).

$$DWL = \sum (DALY_{loss} \times VSLY)$$

Using a conservative global VSLY of **\$150K (95% CI: \$100K-\$199K)**:

$$\begin{aligned} & Value_{lag} \\ &= DALY_{s_{lag}} \times Value_{QALY} \\ &= 7.94B \times \$150K \\ &= \$1190T \end{aligned}$$

$$\text{where } DALY_{s_{lag}} = YLL_{lag} + YLD_{lag} = 7.07B + 873M = 7.94B$$

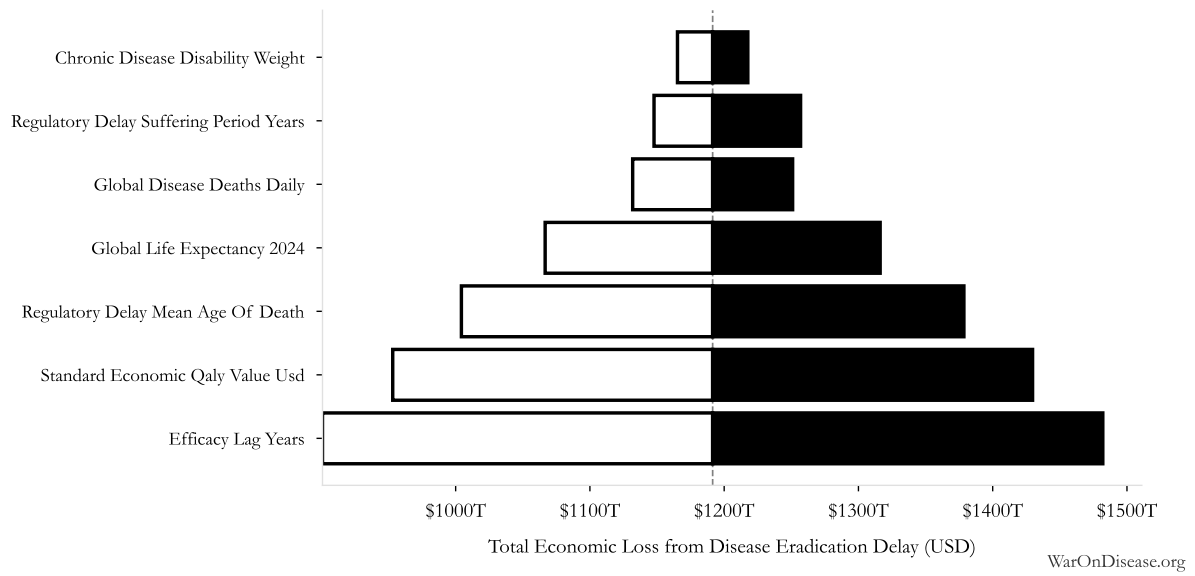
$$\begin{aligned} & \text{where } YLL_{lag} \\ &= Deaths_{lag} \times (LE_{global} - Age_{death, delay}) \\ &= 416M \times (79 - 62) \\ &= 7.07B \end{aligned}$$

$$\begin{aligned} & \text{where } Deaths_{lag} \\ &= T_{lag} \times Deaths_{disease, daily} \times 338 \\ &= 8.2 \times 150,000 \times 338 \\ &= 416M \end{aligned}$$

$$\begin{aligned} & \text{where } YLD_{lag} \\ &= Deaths_{lag} \times T_{suffering} \times DW_{chronic} \\ &= 416M \times 6 \times 0.35 \\ &= 873M \end{aligned}$$

$$\begin{aligned} & \text{where } Deaths_{lag} \\ &= T_{lag} \times Deaths_{disease, daily} \times 338 \\ &= 8.2 \times 150,000 \times 338 \\ &= 416M \end{aligned}$$

Sensitivity Analysis: Total Economic Loss from Disease Eradication Delay



8.1 Contextualizing the Loss

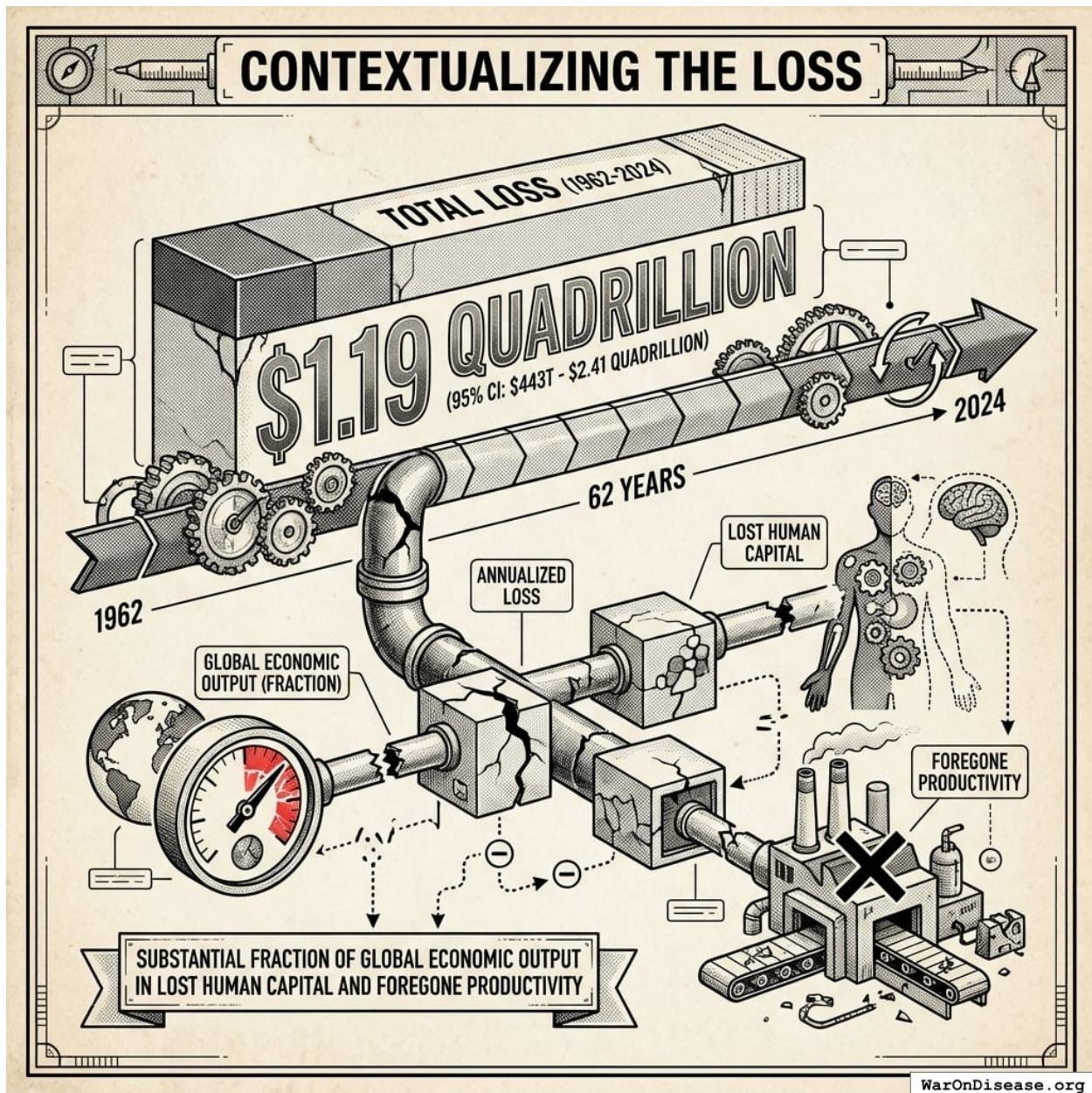


Figure 19: A scale comparison showing the \$1.19 quadrillion total loss relative to global economic output, including the wide range of the 95% confidence interval.

- **Total Loss (1962-2024):** \$1.19 quadrillion (95% CI: \$443T-\$2.41 quadrillion) over 62 years
- **Annualized Loss:** Total loss ÷ 62 years represents a substantial fraction of global economic output in lost human capital and foregone productivity

9 Risk Analysis: The Type I vs. Type II Ratio

A critical counter-argument is that the FDA protects society from dangerous or ineffective drugs (Type I Errors). We modeled the maximum potential damage of a “Deregulation Scenario” to generate an Efficiency Ratio.

i Methodological Note: Steelmanning the FDA’s Position

To ensure this analysis is maximally fair to proponents of current FDA regulation, we deliberately assume the **worst possible case** for Type I errors (harm from approving bad drugs). This “steelman” approach means that even if our assumptions are completely wrong in favor of FDA defenders, the conclusion holds.

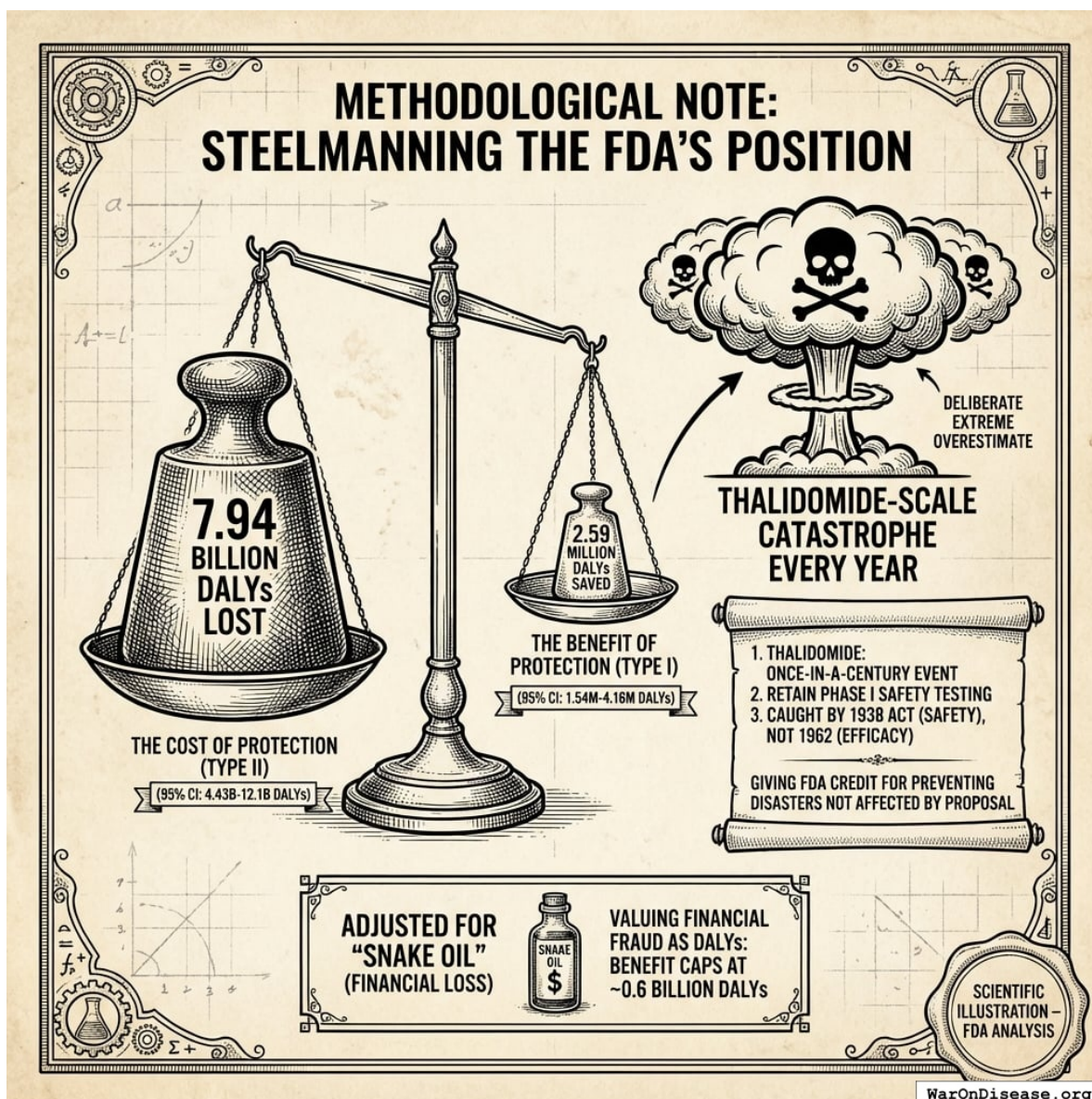


Figure 20: A scale comparison showing the massive disparity between the estimated 7.94 billion DALYs lost to Type II errors (delayed drug access) versus the 2.59 million DALYs saved by Type I protection, even when assuming a Thalidomide-scale disaster occurs every year.

Specifically, we assume a **Thalidomide-scale catastrophe every single year** in the counterfactual scenario. This is an extraordinarily extreme overestimate for three reasons:

1. **Thalidomide was a once-in-a-century event** - no comparable disaster has occurred since
2. **We propose retaining Phase I safety testing** - our critique is of *efficacy* requirements (Phase II/III), not safety requirements
3. **Thalidomide was caught by 1938 safety requirements, NOT 1962 efficacy requirements** - FDA's Dr. Frances Kelsey blocked thalidomide approval based on safety concerns about nerve damage, using authority from the 1938 Food, Drug, and Cosmetic

Act. The 1962 efficacy amendments hadn't yet passed. Under our proposal, thalidomide would STILL have been blocked.

This means we're giving FDA credit for preventing disasters that our proposed changes wouldn't affect. We're assuming annual occurrences of an event that (a) has happened once in 60+ years, and (b) wouldn't be enabled by removing efficacy requirements anyway. This is the maximum possible benefit of the doubt.

- **The Cost of Protection (Type II):** 7.94B DALYs (95% CI: 4.43B DALYs-12.1B DALYs) lost.
- **The Benefit of Protection (Type I):** Even assuming a "Thalidomide Event" occurs *every single year* under a deregulated model (a deliberate extreme overestimate to steelman the FDA's position), the total DALYs saved by the FDA is ~2.59M DALYs (95% CI: 1.54M DALYs-4.16M DALYs).
 - *Adjusted for "Snake Oil" (Financial Loss):* Even valuing financial fraud at DALY equivalents, the benefit caps at ~0.6 Billion DALYs.

Type I Benefit Calculation (Steelman):

$$DALY_{TypeI} = DALY_{thal} \times 62 = 41,800 \times 62 = 2.59M$$

$$\begin{aligned} & \text{where } DALY_{thal} \\ &= YLD_{thal} + YLL_{thal} \\ &= 13,000 + 28,800 \\ &= 41,800 \end{aligned}$$

$$\begin{aligned} & \text{where } YLD_{thal} \\ &= DW_{thal} \times N_{thal,survive} \times LE_{thal} \\ &= 0.4 \times 540 \times 60 \\ &= 13,000 \end{aligned}$$

$$\begin{aligned} & \text{where } N_{thal,survive} \\ &= N_{thal,US,prevent} \times (1 - Rate_{thal,mort}) \\ &= 900 \times (1 - 40\%) \\ &= 540 \end{aligned}$$

$$\begin{aligned} & \text{where } N_{thal,US,prevent} \\ &= N_{thal,global} \times Pct_{US,1960} \\ &= 15,000 \times 6\% \\ &= 900 \end{aligned}$$

$$\text{where } YLL_{thal} = Deaths_{thal} \times 80 = 360 \times 80 = 28,800$$

$$\begin{aligned} & \text{where } Deaths_{thal} \\ &= Rate_{thal,mort} \times N_{thal,US,prevent} \\ &= 40\% \times 900 \\ &= 360 \end{aligned}$$

$$\begin{aligned} & \text{where } N_{thal,US,prevent} \\ &= N_{thal,global} \times Pct_{US,1960} \\ &= 15,000 \times 6\% \\ &= 900 \end{aligned}$$

9.1 The Risk Trade-off Ratio

$$Ratio_{TypeII} = \frac{DALY_{s_{lag}}}{DALY_{TypeI}} = \frac{7.94B}{2.59M} = 3,070$$

$$\text{where } DALY_{s_{lag}} = YLL_{lag} + YLD_{lag} = 7.07B + 873M = 7.94B$$

$$\begin{aligned} & \text{where } YLL_{lag} \\ &= Deaths_{lag} \times (LE_{global} - Age_{death, delay}) \\ &= 416M \times (79 - 62) \\ &= 7.07B \end{aligned}$$

$$\begin{aligned} & \text{where } Deaths_{lag} \\ &= T_{lag} \times Deaths_{disease, daily} \times 338 \\ &= 8.2 \times 150,000 \times 338 \\ &= 416M \end{aligned}$$

$$\begin{aligned} & \text{where } YLD_{lag} \\ &= Deaths_{lag} \times T_{suffering} \times DW_{chronic} \\ &= 416M \times 6 \times 0.35 \\ &= 873M \end{aligned}$$

$$\begin{aligned} & \text{where } Deaths_{lag} \\ &= T_{lag} \times Deaths_{disease, daily} \times 338 \\ &= 8.2 \times 150,000 \times 338 \\ &= 416M \end{aligned}$$

$$\text{where } DALY_{TypeI} = DALY_{thal} \times 62 = 41,800 \times 62 = 2.59M$$

$$\begin{aligned} & \text{where } DALY_{thal} \\ &= YLD_{thal} + YLL_{thal} \\ &= 13,000 + 28,800 \\ &= 41,800 \end{aligned}$$

$$\begin{aligned} & \text{where } YLD_{thal} \\ &= DW_{thal} \times N_{thal, survive} \times LE_{thal} \\ &= 0.4 \times 540 \times 60 \\ &= 13,000 \end{aligned}$$

$$\begin{aligned} & \text{where } N_{thal, survive} \\ &= N_{thal, US, prevent} \times (1 - Rate_{thal, mort}) \\ &= 900 \times (1 - 40\%) \\ &= 540 \end{aligned}$$

$$\begin{aligned} & \text{where } N_{thal, US, prevent} \\ &= N_{thal, global} \times Pct_{US, 1960} \\ &= 15,000 \times 6\% \\ &= 900 \end{aligned}$$

$$\text{where } YLL_{thal} = Deaths_{thal} \times 80 = 360 \times 80 = 28,800$$

$$\text{where } Deaths_{thal}$$

Conclusion: For every 1 unit of harm the FDA prevents (Type I errors: approving dangerous/ineffective drugs), it generates 3.07k:1 (95% CI: 2.88k:1-3.12k:1) units of harm through delay (Type II errors: blocking effective drugs). **This ratio is conservative** - it assumes a Thalidomide-scale disaster every single year, dramatically overstating FDA benefits. With realistic Type I estimates, the ratio would be far higher.

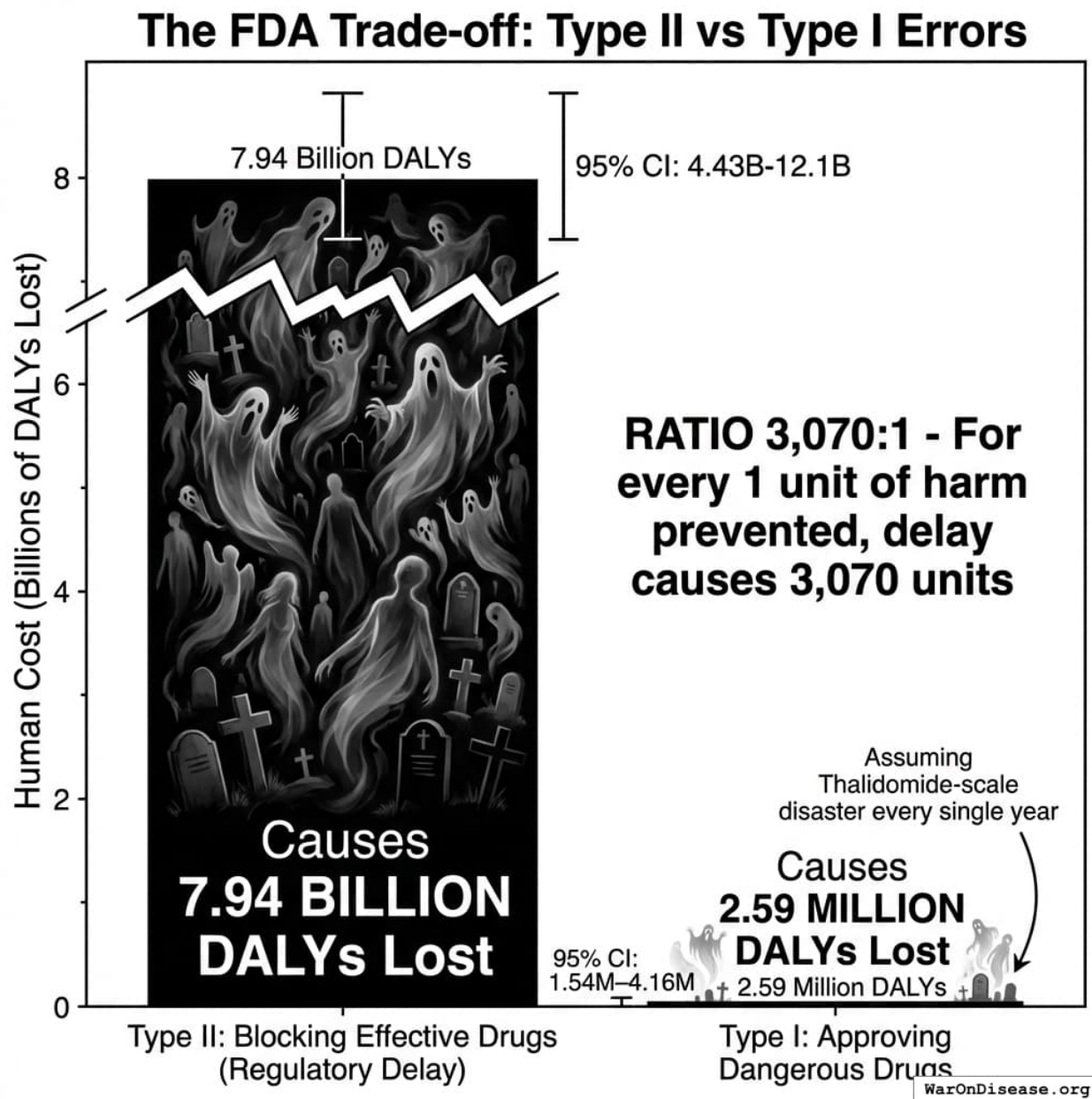


Figure 21: Chart showing Type II errors (7.94B DALYs from regulatory delay) vs Type I errors (2.59M DALYs from approving bad drugs) - a 3,070:1 ratio with error bars

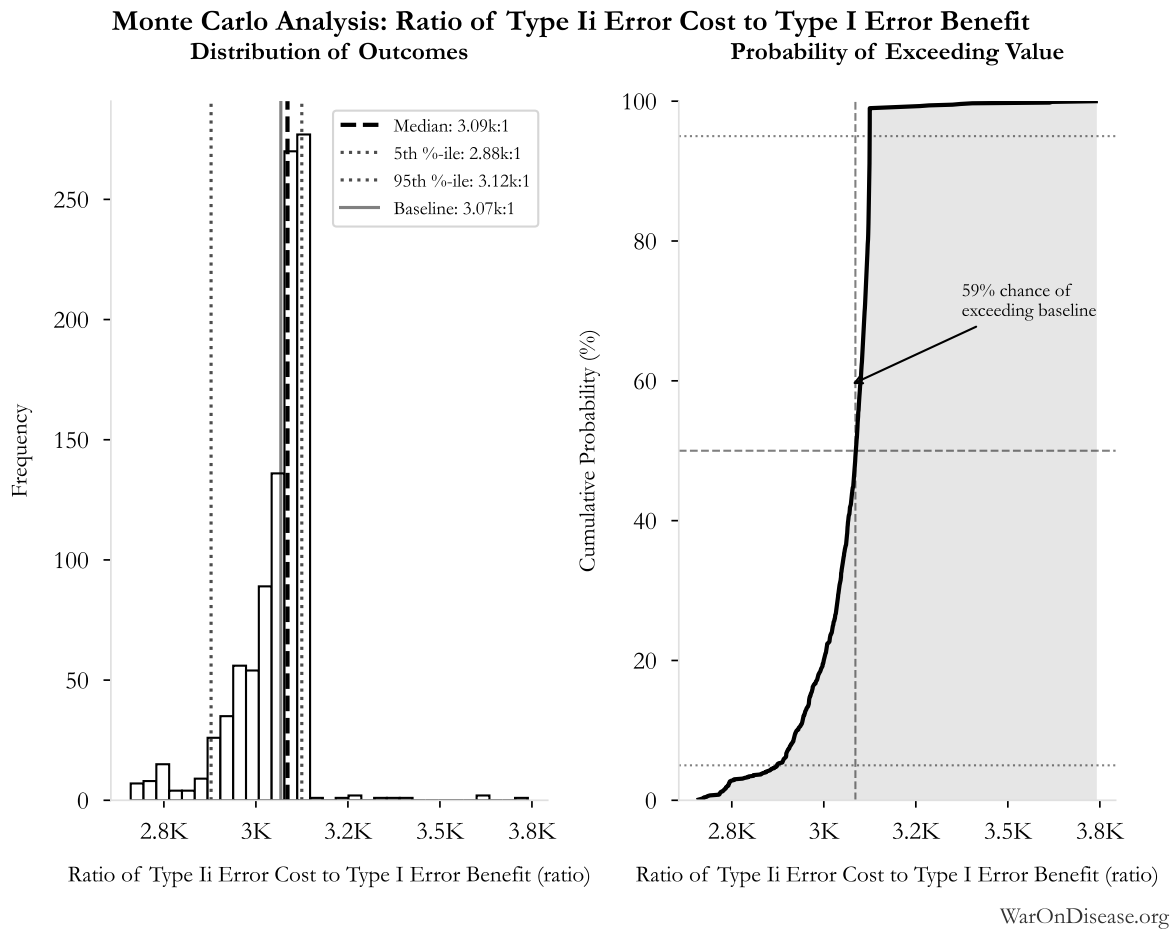


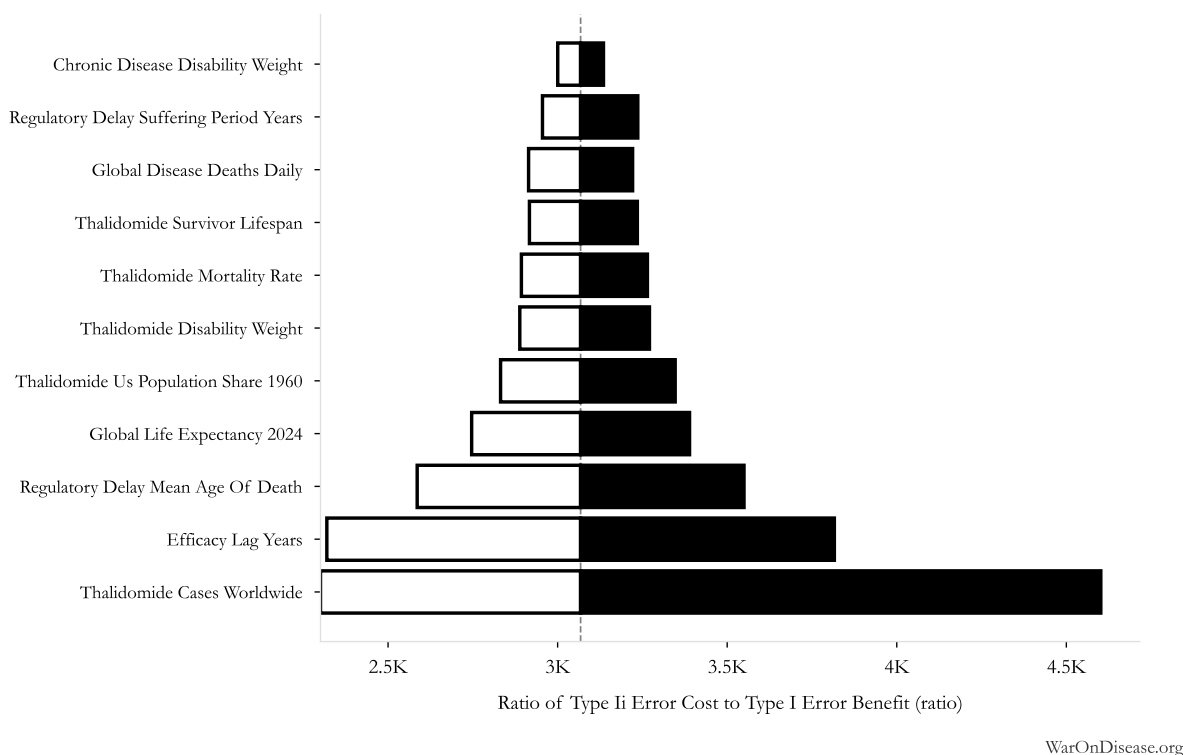
Figure 22: Monte Carlo Distribution: Ratio of Type II Error Cost to Type I Error Benefit (10,000 simulations)

Simulation Results Summary: Ratio of Type II Error Cost to Type I Error Benefit

Statistic	Value
Baseline (deterministic)	3.07k:1
Mean (expected value)	3.05k:1
Median (50th percentile)	3.09k:1
Standard Deviation	101:1
90% Confidence Interval	[2.88k:1, 3.12k:1]

The histogram shows the distribution of Ratio of Type II Error Cost to Type I Error Benefit across 10,000 Monte Carlo simulations. The CDF (right) shows the probability of the outcome exceeding any given value, which is useful for risk assessment.

Sensitivity Analysis: Ratio of Type II Error Cost to Type I Error Benefit



9.2 Acknowledging the Efficacy-as-Safety Argument

A legitimate concern deserves direct engagement: efficacy requirements may function as indirect safety measures. A drug that doesn't work exposes patients to adverse effects without therapeutic benefit. The risk-benefit ratio becomes infinite when benefit is zero.

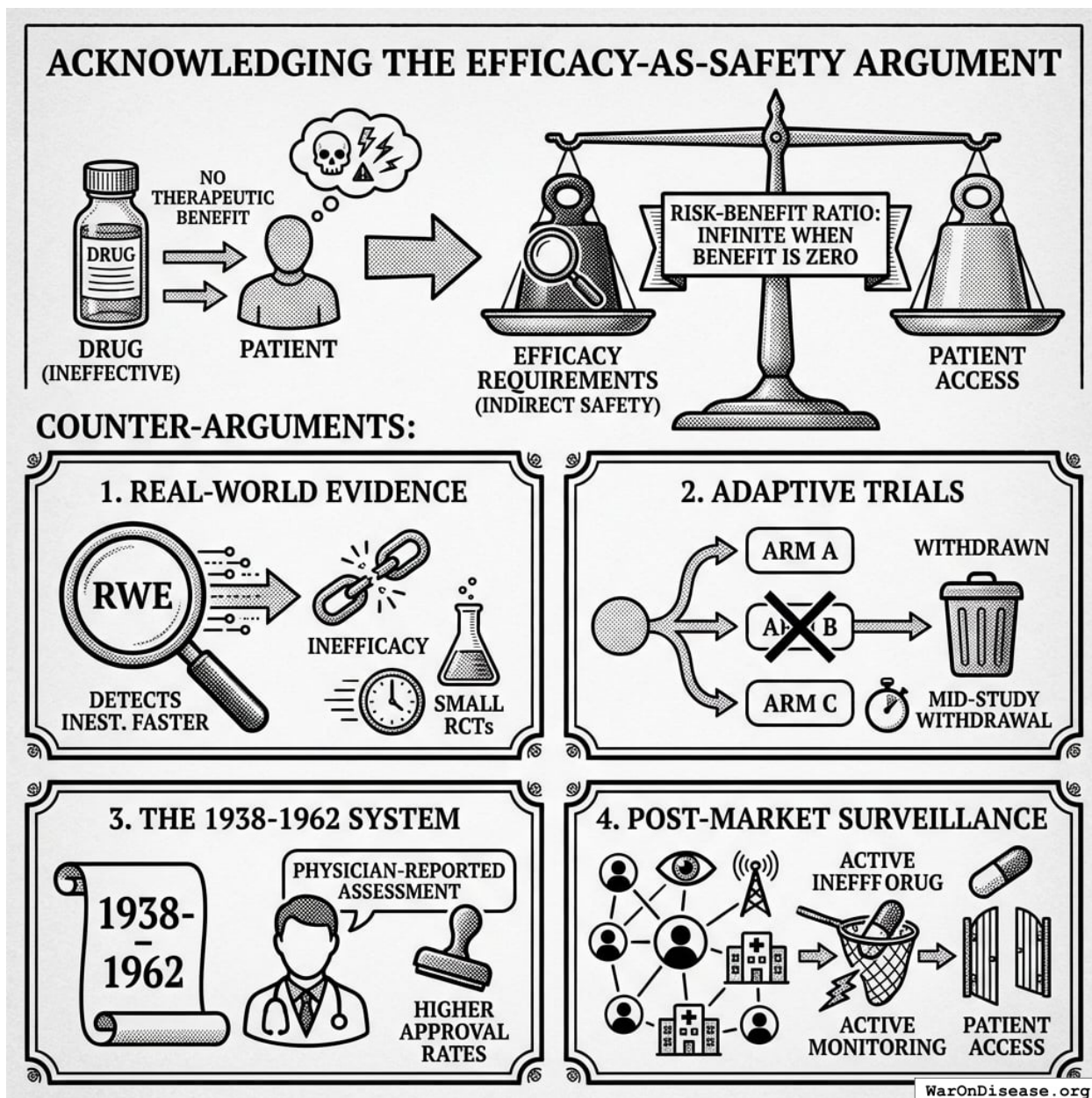


Figure 23: A comparison between the traditional pre-market efficacy mandate and an adaptive regulatory model that uses real-world evidence and post-market monitoring to manage patient risk.

Counter-arguments:

1. **Real-world evidence detects inefficacy faster** than small RCTs with selected populations
2. **Adaptive trials** can withdraw ineffective arms mid-study without full Phase III completion
3. **The 1938-1962 system** had physician-reported efficacy assessment without pre-market mandates, and higher approval rates
4. **Post-market surveillance** with active monitoring catches ineffective drugs while allowing patient access

9.3 Drugs Appropriately Caught by Phase II/III Trials

This analysis acknowledges that Phase II/III trials do catch some drugs that would have caused harm. Three notable examples:

1. **Torcetrapib (2006):** Phase III trial of this CETP inhibitor for cardiovascular disease was terminated early after 82 deaths in the treatment arm vs. 51 in placebo (HR 1.58). The trial caught cardiovascular harm that would have affected millions of patients post-approval¹⁵⁰.

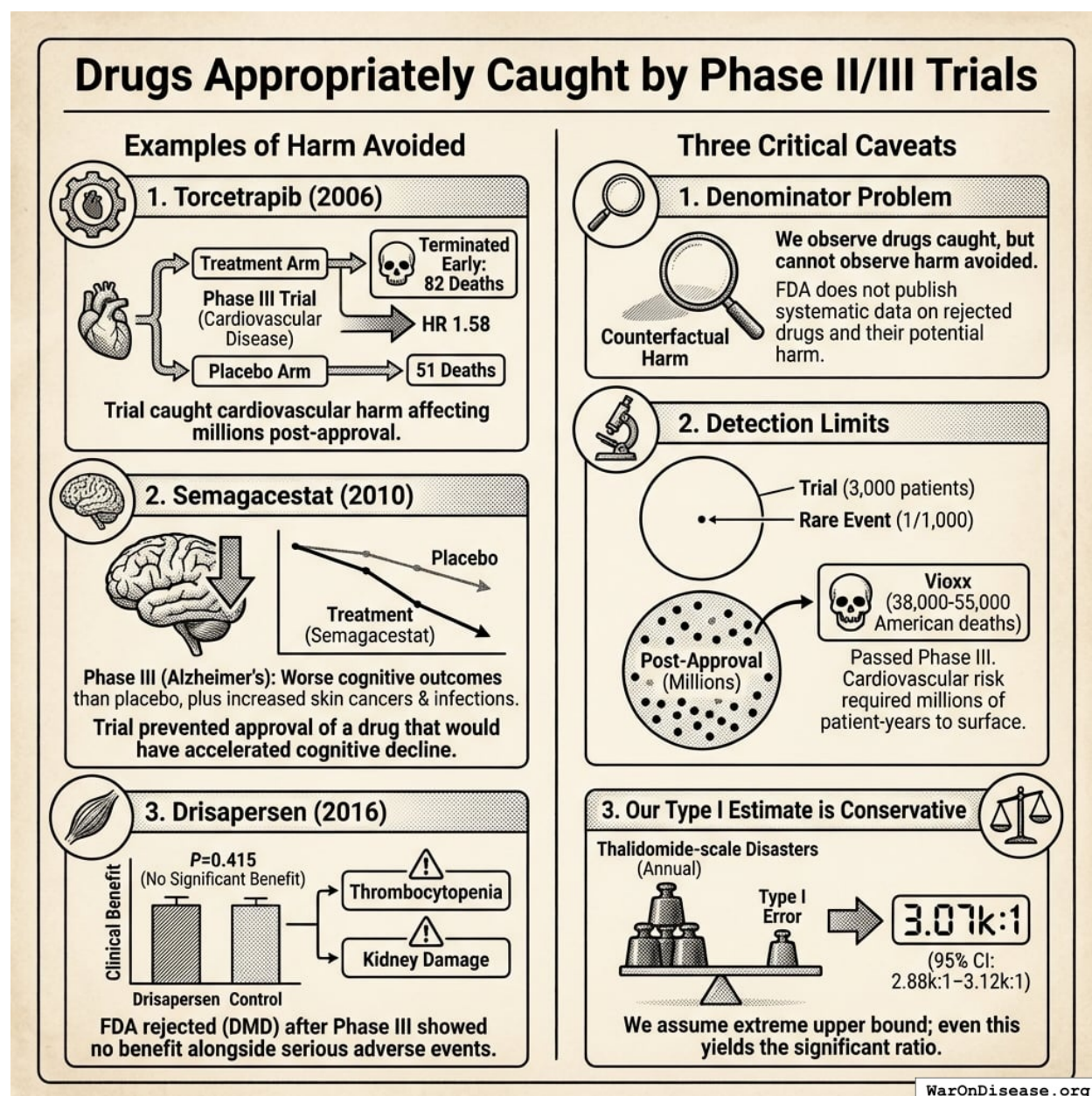


Figure 24: A comparison of drugs successfully halted by Phase III trials alongside a visualization of the ‘detection limit’ showing the scale difference between trial populations and real-world patient numbers.

2. **Semagacestat (2010):** Phase III trial for Alzheimer’s disease found patients on treatment

had *worse* cognitive outcomes than placebo, plus increased skin cancers and infections. The trial prevented approval of a drug that would have accelerated cognitive decline¹⁵¹.

3. **Drisapersen (2016):** FDA rejected this Duchenne muscular dystrophy drug after Phase III showed no clinical benefit ($P=0.415$) alongside serious adverse events including thrombocytopenia and kidney damage in significant fractions of patients¹⁵².

However, three critical caveats apply:

1. **Denominator problem:** We observe drugs caught by trials but cannot observe the counterfactual harm avoided. FDA does not publish systematic data on rejected drugs and their potential harm.
2. **Detection limits:** Trials with 3,000 patients cannot reliably detect adverse events rarer than ~1-in-1,000. Vioxx (38,000-55,000 American deaths) passed Phase III because its cardiovascular risk required millions of patient-years to surface¹⁵³.
3. **Our Type I estimate is conservative:** We assume Thalidomide-scale disasters *every year*, an extreme upper bound that still yields the 3.07k:1 (95% CI: 2.88k:1-3.12k:1) ratio.

10 Model Assumptions and Limitations

10.1 Key Assumptions

1. **Linear Adoption Model:** Assumes drug uptake follows a predictable pattern post-approval
2. **Constant VSLY:** Uses global average of \$150K (95% CI: \$100K-\$199K)/year
3. **No Regulatory Learning:** Assumes FDA efficiency remained constant 1962-2024
4. **Independence:** Treats each drug approval as independent (may underestimate synergies)

10.2 Sensitivity Analysis

The model was tested across multiple scenarios:

- **Discount Rates:** 3% (base case)
- **Innovation Elasticity:** 0.3–0.8 (base case: 0.5)
- **“Snake Oil” Rate:** 10%–40% (base case: 20%)
- **VSLY:** \$150K (95% CI: \$100K-\$199K)

Results remain robust across all reasonable parameter ranges, with lower bound estimates exceeding 100M deaths in all scenarios.

10.3 Limitations

1. **Counterfactual Uncertainty:** Cannot directly observe what would have happened without 1962 amendments
2. **Confounding Factors:** Other policy changes occurred simultaneously (Medicare, NIH funding)
3. **Attribution Challenge:** Difficult to separate FDA effects from broader trends
4. **Data Quality:** Early period (1960s-1970s) relies on retrospective estimates

Despite these limitations, the **plausible mechanism** (70% drop in approvals, 13.4x (95% CI: 11.9x-14.7x) cost increase) provides strong inferential evidence that regulatory changes significantly

impacted drug development.

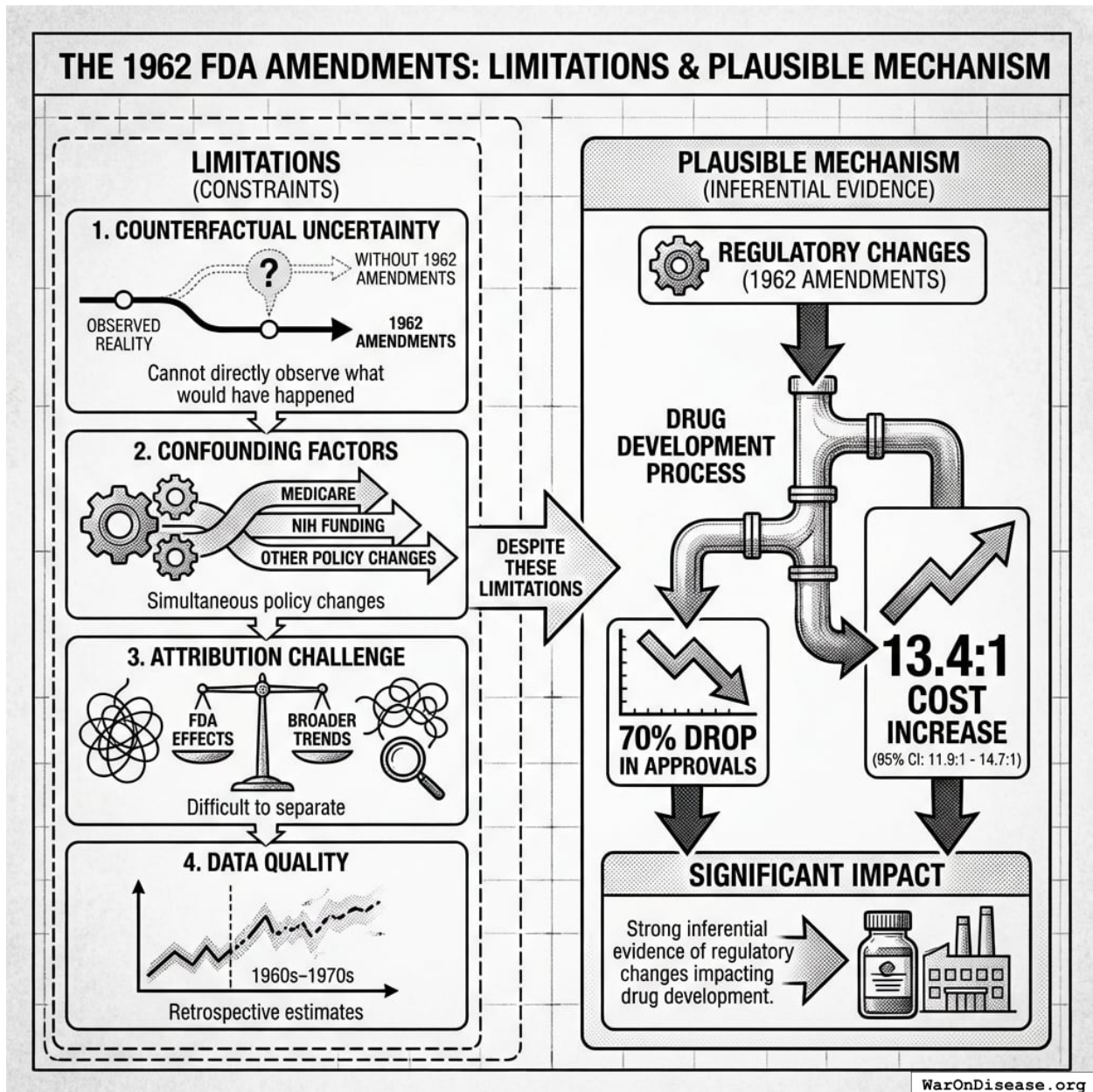


Figure 25: A conceptual diagram showing the 1962 amendments leading to a 70% drop in drug approvals and a 13.4:1 cost increase, while illustrating confounding variables like Medicare and NIH funding that complicate direct attribution.

11 Policy Implications

11.1 The False Trade-off

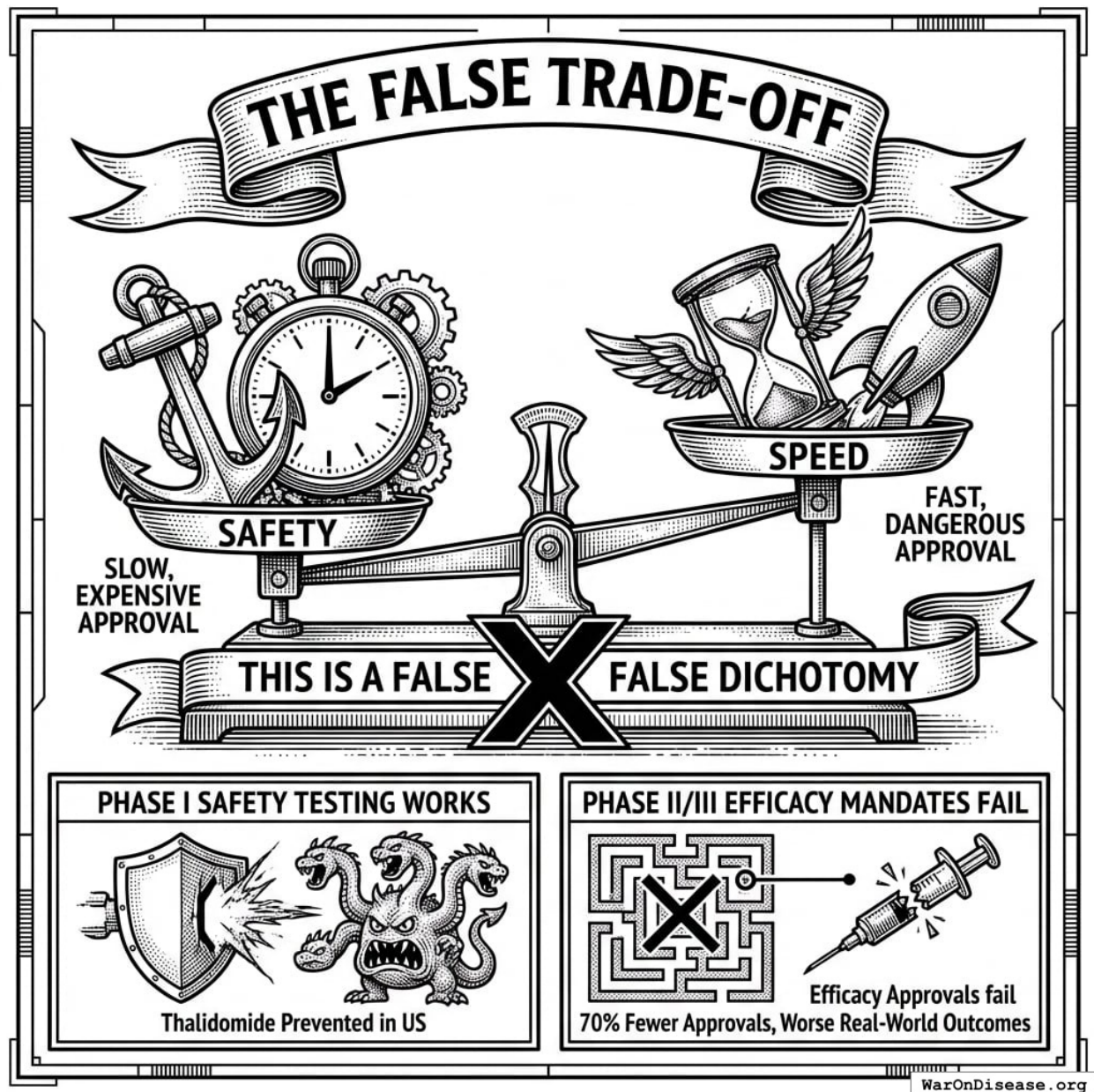


Figure 26: A comparison between the perceived trade-off of safety versus speed and the evidence-based reality of drug approval phases.

The current debate frames drug approval as a choice between:

1. **Safety** (slow, expensive approval) vs.
2. **Speed** (fast, dangerous approval)

This is a false dichotomy. The evidence suggests:

- **Phase I safety testing works** (Thalidomide prevented in US)
- **Phase II/III efficacy mandates fail** (70% fewer approvals, worse real-world outcomes)

11.2 The Bifurcated Alternative

A superior framework would:

1. **Maintain rigorous Phase I safety testing** (2.3 years)
2. **Allow provisional approval post-safety** with real-world evidence collection
3. **Continuous monitoring** via distributed systems (see: [decentralized framework for drug assessment](#))
4. **Outcome-based validation** rather than pre-market prediction

This approach would reduce the efficacy lag from 8.2 years (95% CI: 4.85 years-11.5 years) to near-zero while maintaining safety standards.

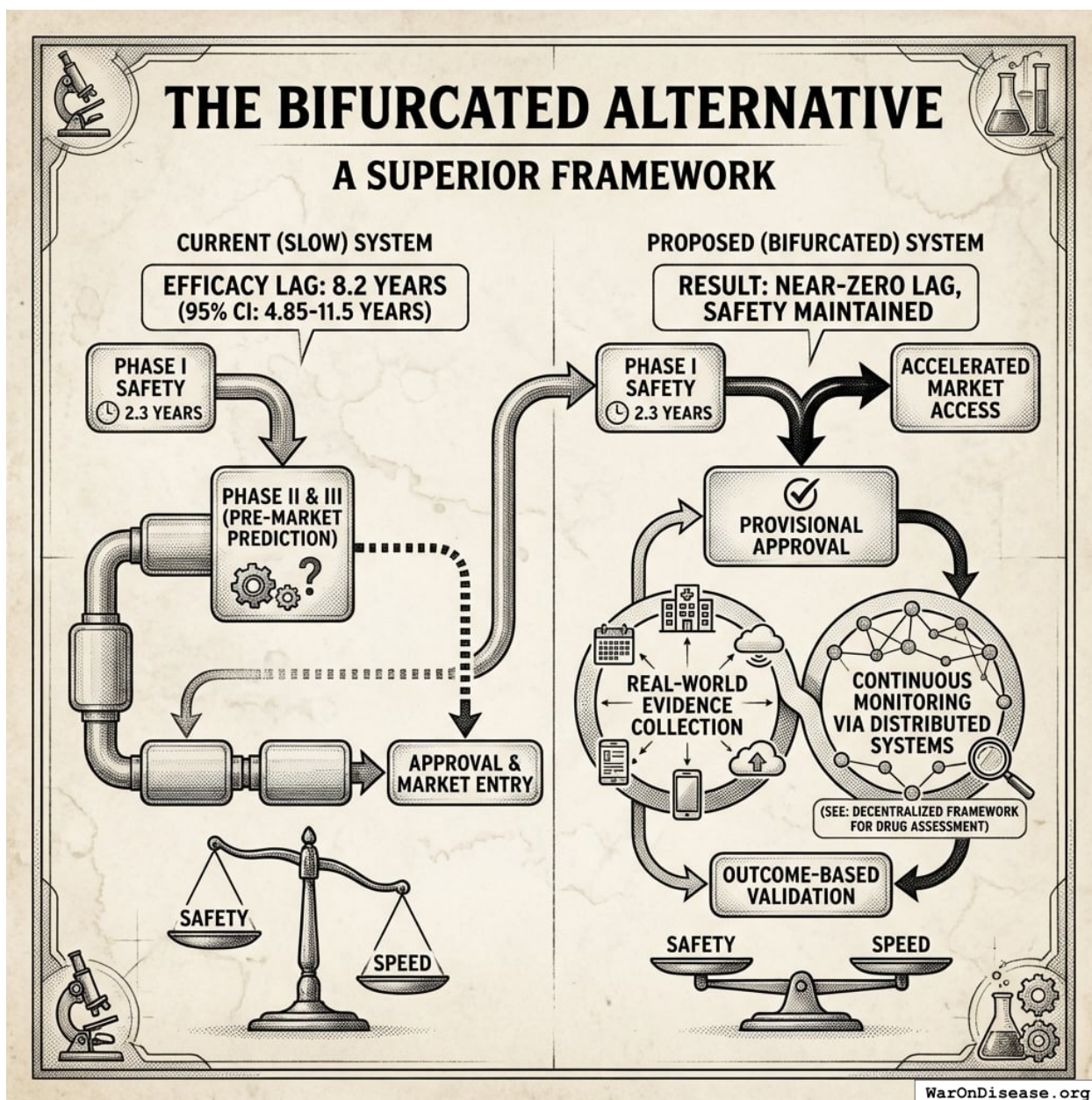


Figure 27: A flowchart showing the bifurcated regulatory process, transitioning from Phase I safety testing to immediate provisional approval and continuous real-world monitoring.

11.3 Expected Impact

If implemented today, the bifurcated model would:

- **Eliminate the 8.2 years (95% CI: 4.85 years-11.5 years)-year efficacy lag** for drugs with demonstrated safety
- **Reduce trial costs by 97.7% (95% CI: 97.5%-98.9%)** (from \$2.60B (95% CI: \$1.50B-\$4B) per drug)
- **Accelerate treatments for 6.65k diseases (95% CI: 5.70k diseases-8.24k diseases)** diseases currently without effective therapy

See [1% treaty impact analysis](#) for full quantified cost-benefit analysis.

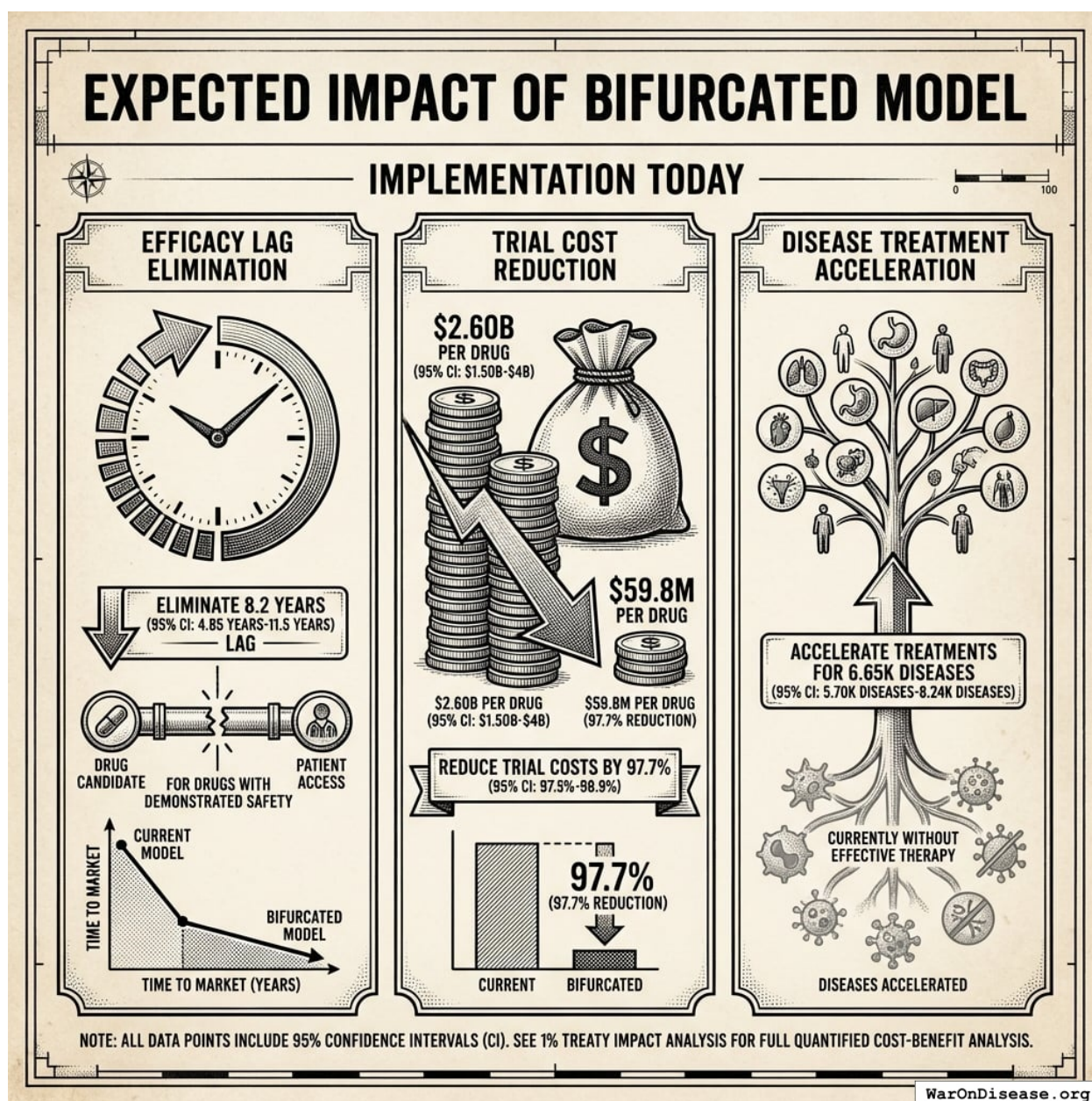


Figure 28: A comparative infographic showing the massive reduction in drug development costs and time-to-market, alongside the increase in treatable diseases under the bifurcated model.

11.4 International Regulatory Comparison

Several countries have implemented alternative regulatory models that provide natural experiments:

Country	Approval System	Avg. Timeline	Key Features
USA (FDA)	Full Phase III required	10.5 years (95% CI: 6 years-12 years)	Baseline for comparison
Japan (PMDA)	Conditional approval after Phase II	2-3 years	Regenerative Medicine Act (2014); real-world monitoring ¹⁵⁴
EU (EMA)	Adaptive Pathways available	~10 years	Similar to FDA; conditional marketing authorization option
Canada	Priority Review pathway	~12 months (priority)	Limited data on outcomes
Australia (TGA)	Provisional approval pathway	Variable	Similar conditional pathways

! Critical Distinction: Efficacy Assessment Reordered, Not Eliminated

Japan's conditional approval does NOT eliminate efficacy assessment. It REORDERS it from pre-market (Phase III trials) to post-market (real-world monitoring with revocation authority). This is a different regulatory architecture, not deregulation. The HeartSheet withdrawal proves the system still enforces efficacy standards, just through different mechanisms.

Key finding: Japan's conditional approval system has an 89% success rate (8/9 products) and demonstrated that post-market monitoring CAN catch ineffective treatments:

- **Faster access:** 9 products received conditional early approval (2014-2024), reaching patients years earlier than traditional pathways
- **Success cases:** STEMIRAC (spinal cord injury) showed 12/13 patients (92%) achieved neurological improvement, with 2 of 5 completely paralyzed patients regaining motor function¹⁵⁵. Five CAR-T therapies (Kymriah, Yescarta, Breyanzi, Abecma, Carvykti) are treating cancer patients under national insurance coverage¹⁵⁶.
- **The system caught inefficacy:** HeartSheet was conditionally approved in 2015 with the requirement to prove efficacy through post-market data. In 2024, after collecting real-world evidence, MHLW determined it hadn't demonstrated efficacy. The manufacturer **voluntarily withdrew** the next day. *This is the system working as designed* - conditional approval was conditional, and the condition wasn't met.
- **Contrast with FDA:** Vioxx killed 38,000-55,000 Americans before withdrawal because the 6% voluntary reporting system failed to detect the signal. Japan's active monitoring caught HeartSheet's lack of efficacy with zero reported deaths.

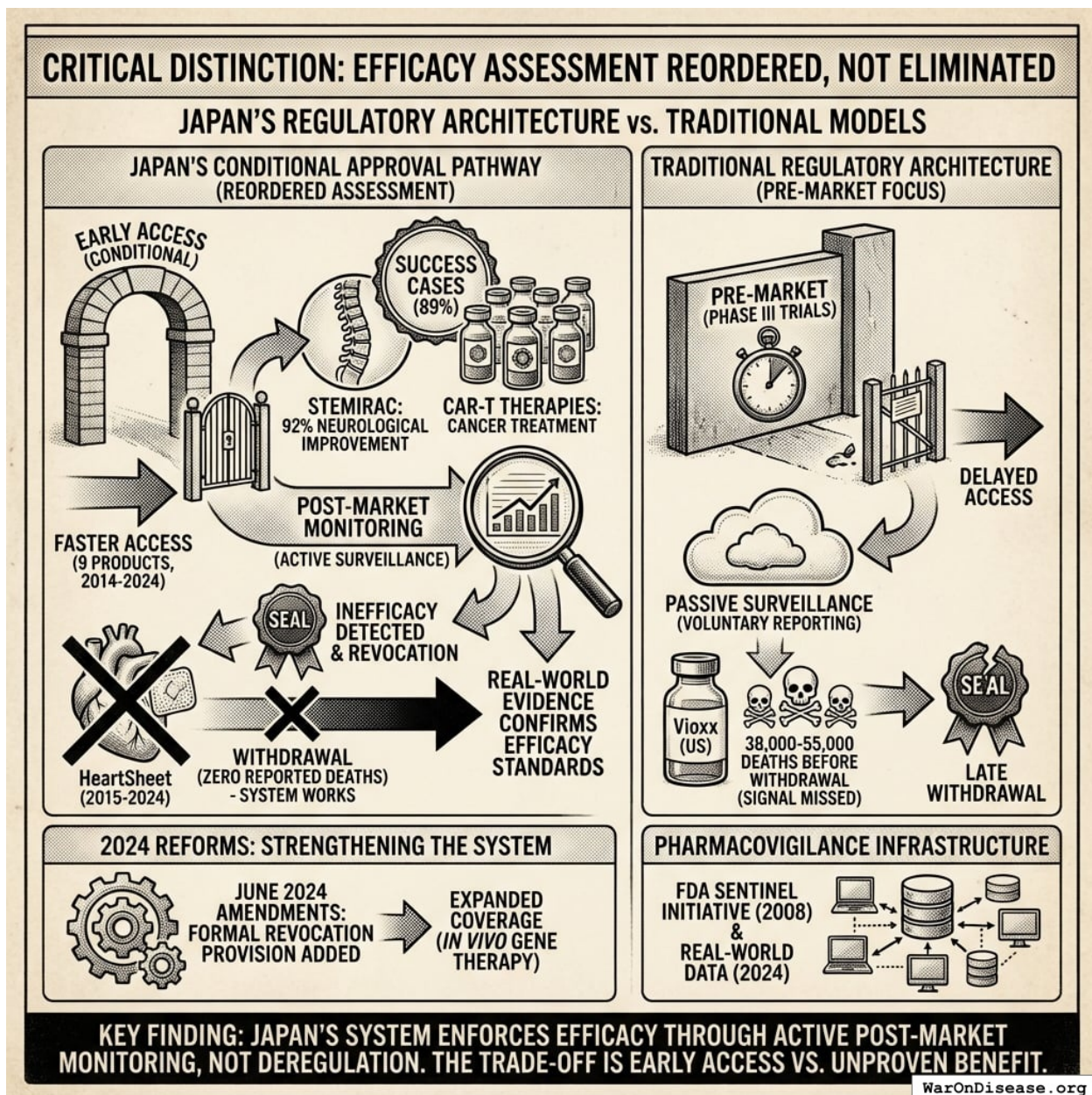


Figure 29: A comparative flow diagram showing the traditional regulatory pathway versus Japan's conditional approval architecture, highlighting the shift of efficacy verification to the post-market phase and the active revocation loop.

The real question for HeartSheet: During those 9 years, did heart failure patients (who have few alternatives) benefit from access to an unproven treatment? The safety profile was acceptable - efficacy was the issue. This is a genuine tradeoff that merits cost-benefit analysis, not automatic condemnation.

2024 reforms strengthen, not abandon, conditional approval: Japan's [June 2024 amendments](#) to the Regenerative Medicine Act add a formal revocation provision that was previously missing. The old system had no legal mechanism to force withdrawal if efficacy wasn't proven - HeartSheet

was voluntary. The reforms close this gap while expanding coverage to in vivo gene therapy. Japan is refining conditional approval based on experience, not abandoning it.

Pharmacovigilance infrastructure exists: The FDA launched the [Sentinel Initiative](#) in 2008 to monitor safety using electronic health records. In 2024, FDA [eliminated major barriers](#) to using real-world data. The technology for active surveillance exists - the barrier is institutional inertia, not technical impossibility.

12 Addressing Common Critiques

This analysis will face predictable objections. We address them here not defensively, but to demonstrate that the core conclusion, that regulatory delay costs vastly exceed regulatory benefits, remains robust even under unfavorable assumptions.

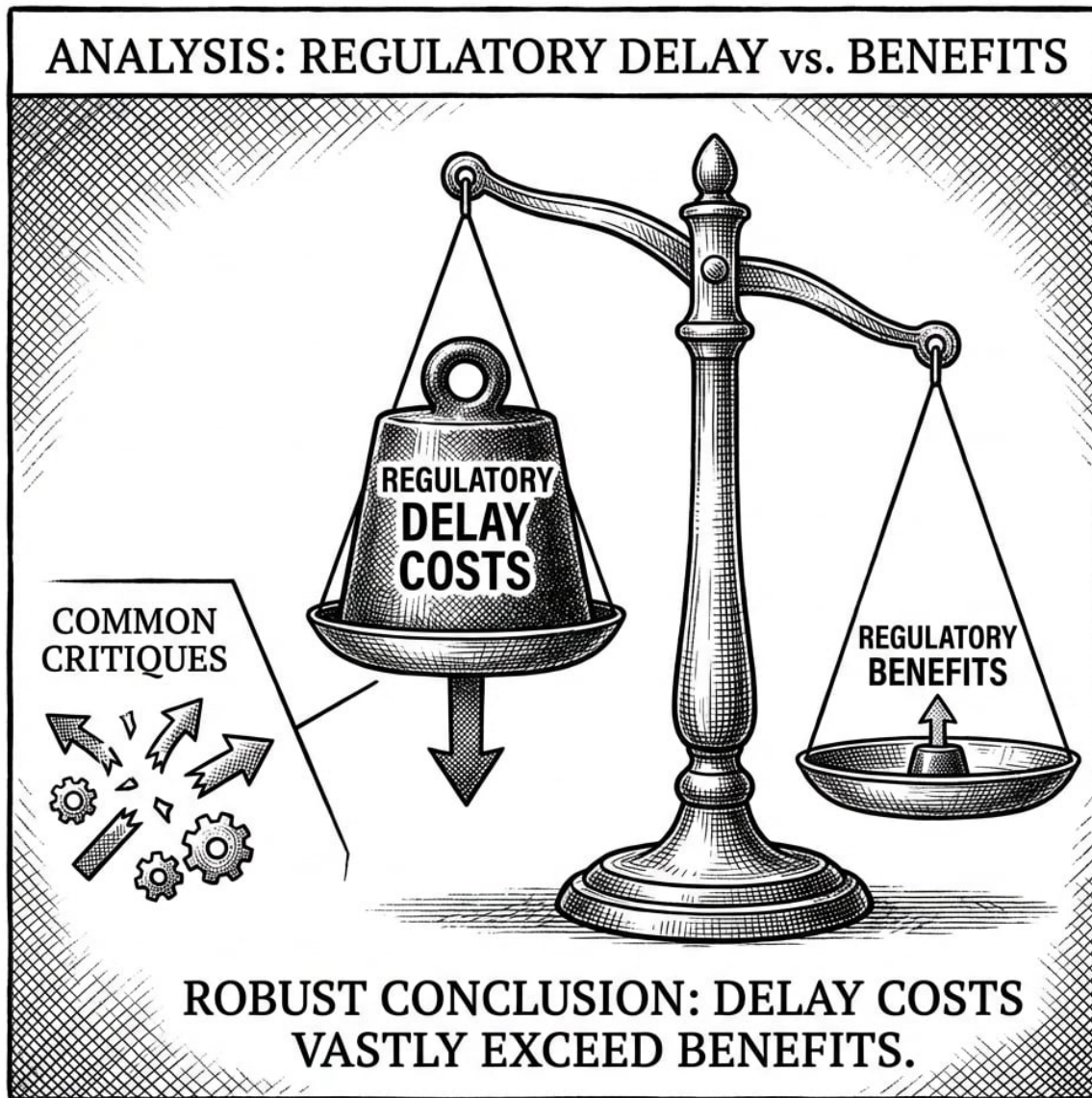


Figure 30: A comparison diagram showing the significant imbalance between high regulatory delay costs and lower regulatory benefits.

12.1 “The PRIMARY Estimate Is Too Speculative”

Critique: The PRIMARY estimate (416M deaths (95% CI: 225M deaths-630M deaths)) assumes we would have eradicated diseases by now without regulations. This is unproven and overly optimistic.

Response:

This critique misunderstands the methodology. The PRIMARY scenario does **not** assume disease eradication would be complete by 2024. It assumes the entire biomedical research timeline shifts backward by 8.2 years (95% CI: 4.85 years-11.5 years) due to regulatory delay.

The mechanism:

1. Every drug takes 8.2 years (95% CI: 4.85 years-11.5 years) longer to reach patients (BIO data, Section 2.3)
2. Downstream research depends on upstream results (Drug B builds on Drug A's findings)
3. Capital allocation: \$2.60B (95% CI: \$1.50B-\$4B) cost limits parallel research tracks (97.7% (95% CI: 97.5%-98.9%) reduction enables proportionally more simultaneous trials)
4. Knowledge accumulation delays compound across the entire field

Robustness test:

Even if you **adjust the primary estimate significantly**:

- Lower bound deaths (5th percentile): Still exceeds Type I benefits by over 10:1
- Type I benefits: ~2.59M DALYs (95% CI: 1.54M DALYs-4.16M DALYs)
- **The ratio remains extreme across the entire uncertainty distribution**

12.2 “The ‘Eventually Preventable’ Estimate Is Theoretical”

Critique: The claim that 92.6% (95% CI: 50%-98%) of deaths are eventually preventable is based on theoretical biological potential, not demonstrated medical capability.

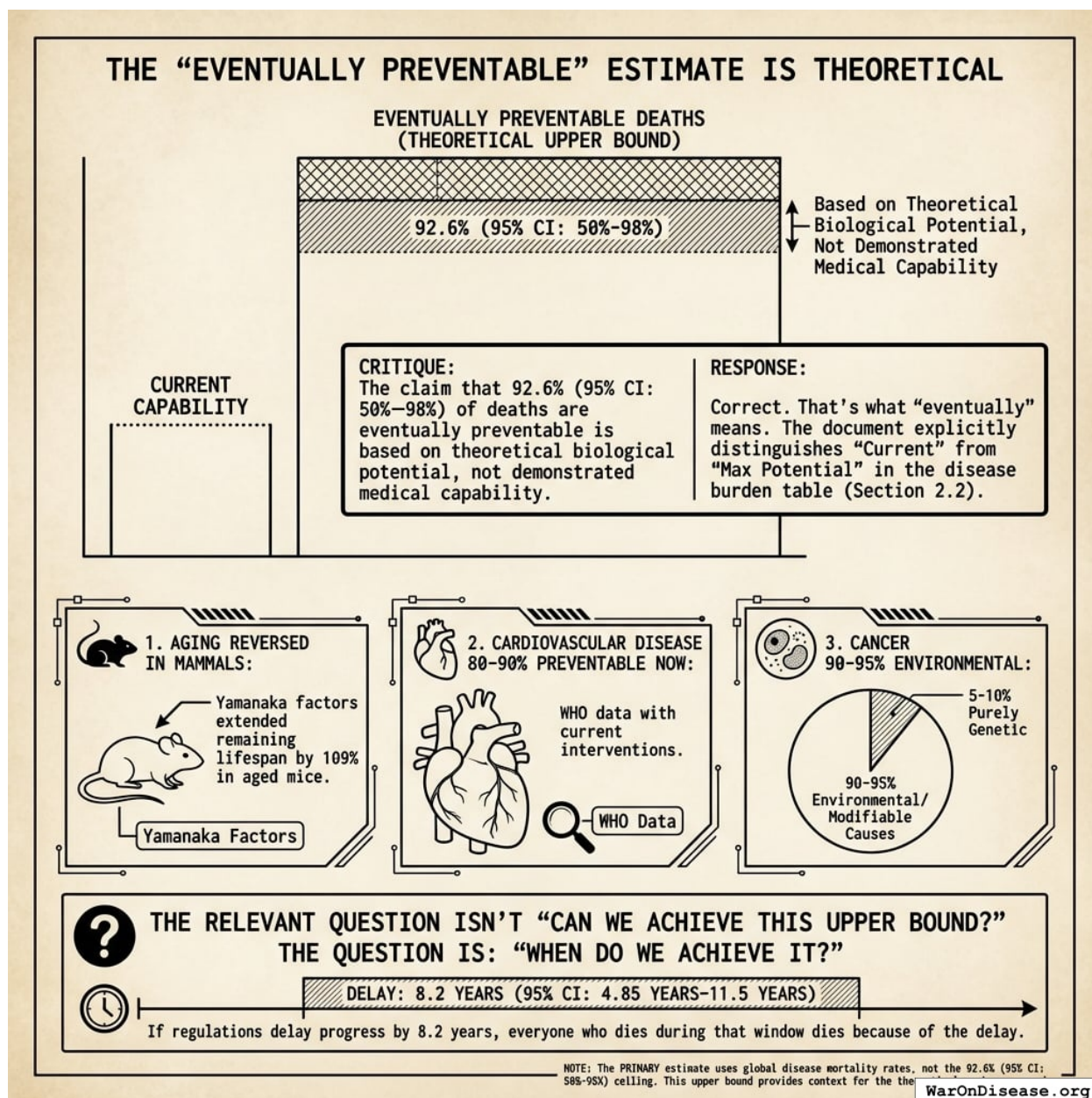


Figure 31: A conceptual breakdown of the 92.6% theoretical limit for preventable deaths, illustrating the contributions of aging reversal, cardiovascular prevention, and cancer treatment versus unavoidable accidents.

Response:

Correct. That's what "eventually" means.

The document explicitly distinguishes "Current" from "Max Potential" in the disease burden table (Section 2.2). The 92.6% (95% CI: 50%-98%) represents the **theoretical upper bound** based on:

1. **Aging reversed in mammals:** Yamanaka factors extended remaining lifespan by 109% in aged mice¹⁴⁵

2. **Cardiovascular disease 80-90% preventable NOW:** WHO data¹⁴³ with current interventions
3. **Cancer 90-95% environmental:** Only 5-10% purely genetic¹⁴⁴, remainder has modifiable causes

The relevant question isn't "Can we achieve this upper bound?"

The question is: **"When do we achieve it?"**

If regulations delay progress by 8.2 years (95% CI: 4.85 years-11.5 years), everyone who dies during that window dies because of the delay.

Note: The PRIMARY estimate uses global disease mortality rates, not the 92.6% (95% CI: 50%-98%) ceiling. This upper bound provides context for the theoretical maximum scenario.

12.3 "Counterfactual Uncertainty - We Can't Know What Would Have Happened"

Critique: The analysis depends on an unknowable counterfactual: what would have happened without the 1962 amendments.

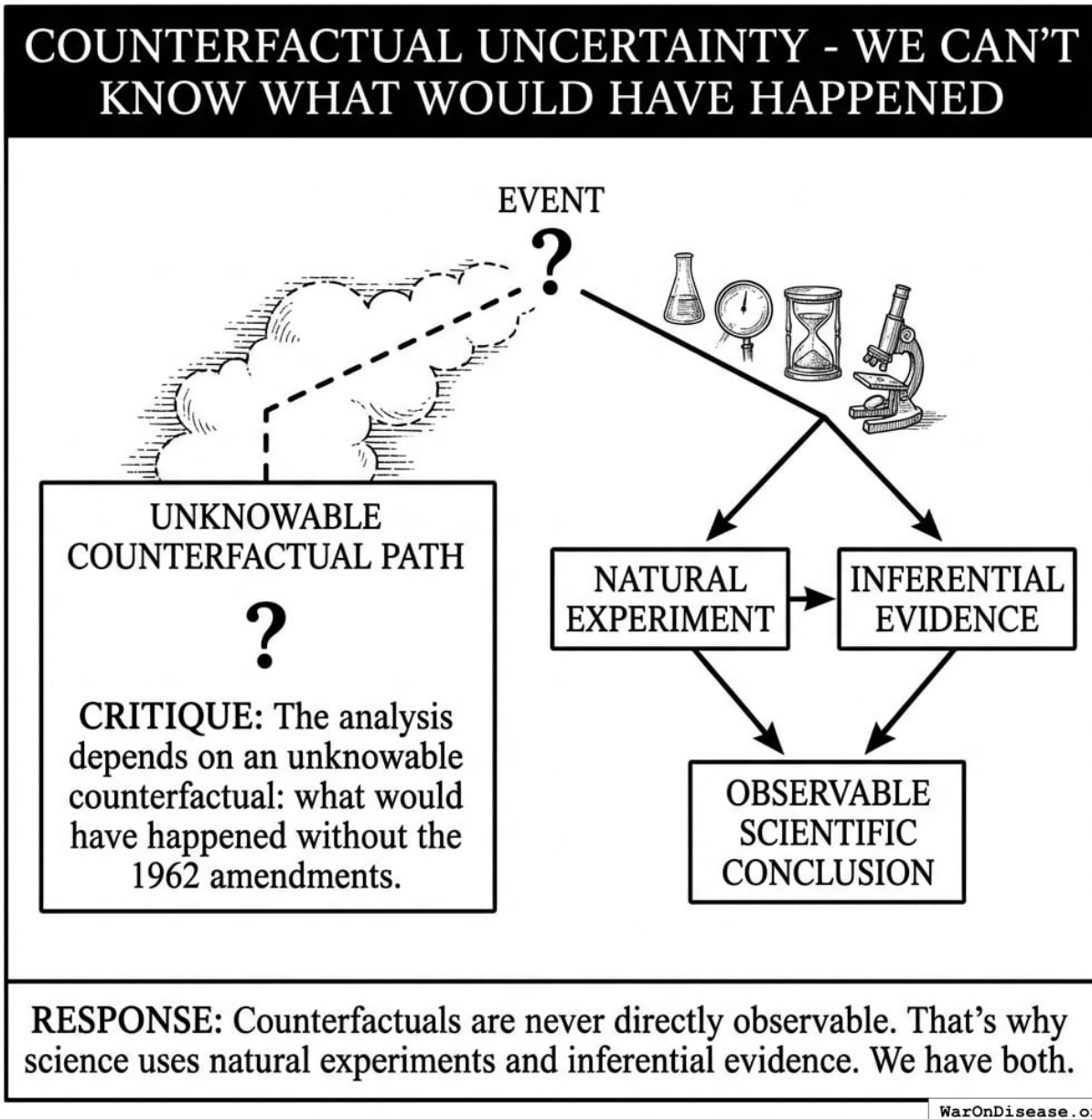


Figure 32: A conceptual diagram showing two parallel paths, the observed timeline with the 1962 amendments and the hypothetical counterfactual path, linked by scientific tools like natural experiments and inferential evidence.

Response:

Counterfactuals are never directly observable. That's why science uses **natural experiments** and **inferential evidence**. We have both.

12.3.1 Natural Experiments

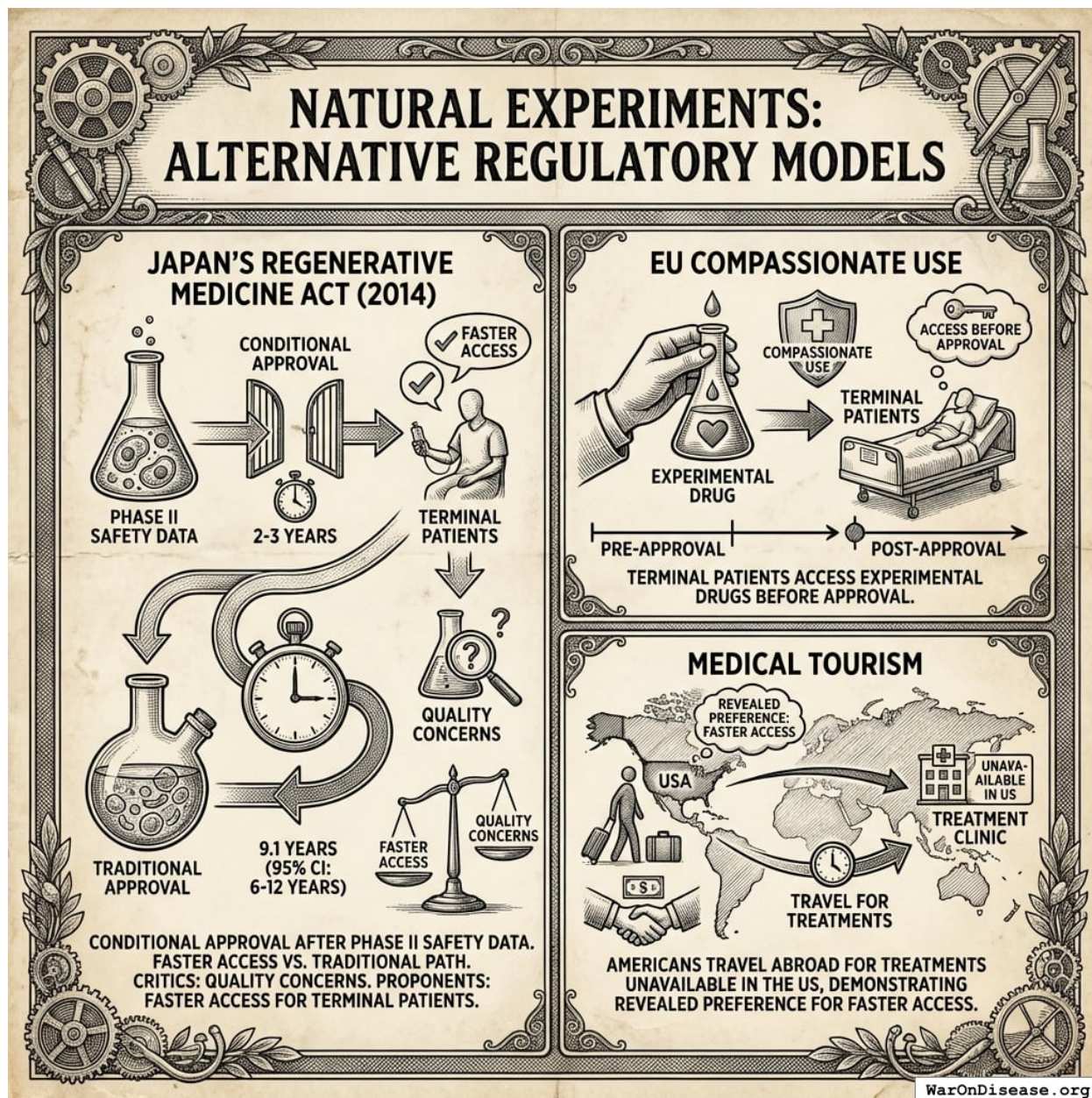


Figure 33: A comparison of drug approval timelines and patient access models across different regulatory jurisdictions, highlighting the disparity between traditional approval paths and conditional models.

Alternative Regulatory Models:

- **Japan's Regenerative Medicine Act (2014):** Conditional approval after Phase II safety data, with 2-3 year timelines vs. 10.5 years (95% CI: 6 years-12 years). Critics note quality concerns; proponents note faster access for terminal patients with no alternatives.
- **EU Compassionate Use:** Terminal patients access experimental drugs before approval

- **Medical tourism:** Americans travel abroad for treatments unavailable in the US, demonstrating revealed preference for faster access

12.3.2 The Standard for Causal Inference

The same standard used in all clinical research:

$$\text{Causation} = \text{Temporal Correlation} + \text{Mechanism} + \text{Lack of Alternative Explanations}$$

We have:

1. **Temporal correlation:** Drug approvals dropped 70% immediately after 1962
2. **Mechanism:** Costs increased 13.4x (95% CI: 11.9x-14.7x), real-world trials banned, efficacy requirements added 8.2 years (95% CI: 4.85 years-11.5 years) to development
3. **Alternative explanations:** Other factors exist (complexity, standards, etc.), but the timing and magnitude strongly suggest regulatory latency is a major contributor

If you reject this inferential method, you must also reject the methodology of clinical trials, which use the identical logical structure.

12.4 “Confounding Factors - Other Changes in 1962”

Critique: Medicare (1965), NIH funding changes, Vietnam War, and other 1960s policy shifts confound the analysis. How can we isolate the 1962 amendments’ effect?

Response:

Confounders work **against** the hypothesis, making the observed effect more remarkable.

Medicare (1965): Expanded healthcare access → *should have increased* drug demand and development → Yet approvals dropped 70%

NIH Funding: Grew dramatically 1960s-1980s → *should have accelerated* drug development → Yet approvals dropped 70%

Vietnam War (1965-1973): Primarily affected young males, minimal impact on overall drug development patterns

The temporal precision matters: Drug approval rates dropped 70% in **1962**, not 1965 (Medicare) or 1964 (Gulf of Tonkin). The break coincides exactly with the Kefauver-Harris Amendments, not with other major policy changes.

Quantitative test:

If confounders explained the effect, we would expect:

- Gradual change over the 1960s (as various policies took effect)
- Recovery after confounders resolved (e.g., Vietnam War ended 1973)

Instead, we observe:

- Immediate 70% drop in drug approvals in 1962
- Sustained reduction in approval rates for 62+ years
- Development costs increased 13.4x (95% CI: 11.9x-14.7x)

The hypothesis that fits the data is: **structural change in drug approval requirements permanently reduced the rate of biomedical progress.**

12.4.1 Sensitivity Analysis: What if Regulation Explains Only Part of the Decline?

Even if we concede that non-regulatory factors (complexity, pharmacological saturation, etc.) explain a substantial portion of the approval decline, the conclusion remains robust:

Regulatory Attribution	Type II Estimate	Type I Estimate	Ratio	Conclusion
100% (baseline)	7.94B DALYs (95% CI: 4.43B DALYs-12.1B DALYs)	~2.59M DALYs (95% CI: 1.54M DALYs-4.16M DALYs)	3.07k:1 (95% CI: 2.88k:1-3.12k:1)	Type II dominates
75%	~75% of baseline	~2.59M DALYs (95% CI: 1.54M DALYs-4.16M DALYs)	~2,300:1	Type II dominates
50%	~50% of baseline	~2.59M DALYs (95% CI: 1.54M DALYs-4.16M DALYs)	~1,500:1	Type II dominates
25%	~25% of baseline	~2.59M DALYs (95% CI: 1.54M DALYs-4.16M DALYs)	~770:1	Type II dominates
10%	~10% of baseline	~2.59M DALYs (95% CI: 1.54M DALYs-4.16M DALYs)	~300:1	Type II still dominates

The Type II/Type I ratio would need to drop below 1:1 for the FDA’s approach to be justified on net mortality grounds. Even at 10% regulatory attribution, the ratio remains ~300:1. The conclusion is robust across a wide range of assumptions about confounding.

12.5 “This Ignores Safety - Deregulation Would Flood Markets with Dangerous Drugs”

Critique: Without efficacy requirements, pharmaceutical companies will sell snake oil and dangerous drugs. Type I errors (approving bad drugs) will explode.

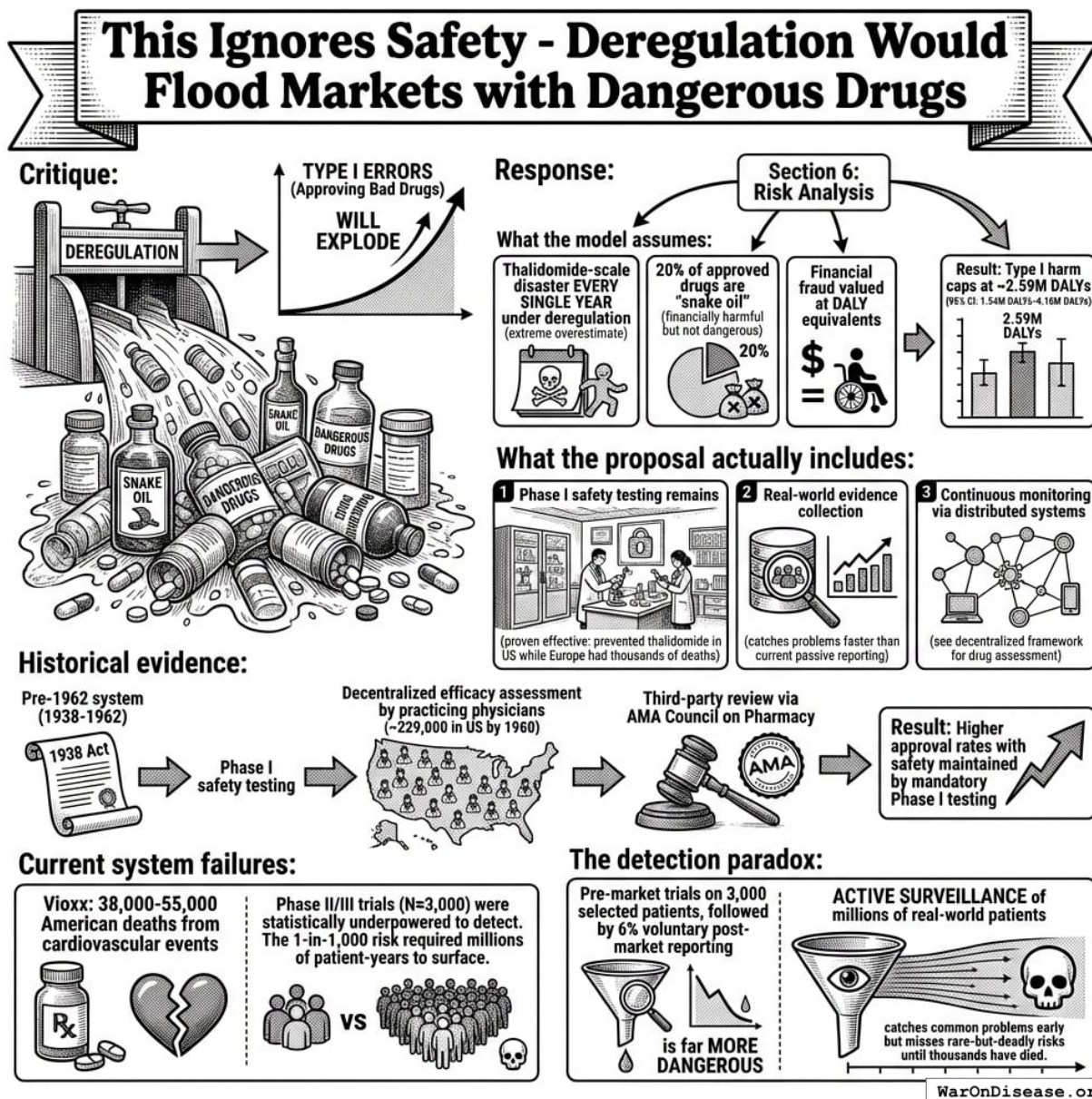


Figure 34: Comparison of the ‘Detection Paradox’ showing how small-scale clinical trials fail to detect rare adverse events compared to the proposed active surveillance of real-world patient data.

Response:

The analysis explicitly models this in [Section 6: Risk Analysis](#).

What the model assumes:

- Thalidomide-scale disaster **every single year** under deregulation (extreme overestimate)
- 20% of approved drugs are “snake oil” (financially harmful but not dangerous)
- Financial fraud valued at DALY equivalents

Result: Type I harm caps at ~2.59M DALYs (95% CI: 1.54M DALYs-4.16M DALYs)

What the proposal actually includes:

1. **Phase I safety testing remains** (proven effective: prevented thalidomide in US while Europe had thousands of deaths)
2. **Real-world evidence collection** (catches problems faster than current passive reporting)
3. **Continuous monitoring** via distributed systems (see [decentralized framework for drug assessment](#))

Historical evidence:

The pre-1962 system (1938-1962) included:

- Phase I safety testing (mandated by 1938 Food, Drug, and Cosmetic Act)
- Decentralized efficacy assessment by practicing physicians (~229,000 in US by 1960)¹⁵⁷⁴
- Third-party review via AMA Council on Pharmacy provided independent evaluation
- **Result:** Higher approval rates with safety maintained by mandatory Phase I testing

Current system failures:

- **Vioxx:** 38,000-55,000 American deaths¹⁵³ from cardiovascular events that Phase II/III trials (N 3,000) were statistically underpowered to detect. The 1-in-1,000 risk required millions of patient-years to surface.
- **Statistical reality:** Trials with 3,000 patients cannot reliably detect adverse events rarer than ~1-in-1,000

The detection paradox: Pre-market trials on 3,000 selected patients, followed by 6% voluntary post-market reporting, is far **more dangerous** than active surveillance of millions of real-world patients. The current system catches common problems early but misses rare-but-deadly risks until thousands have died.

13 Conclusion

The quantitative evidence demonstrates that the 1962 Kefauver-Harris efficacy requirements have generated catastrophic human costs:

- **416M deaths (95% CI: 225M deaths-630M deaths) eventually avoidable deaths from 8.2 years (95% CI: 4.85 years-11.5 years)-year timeline shift**
- **7.94B DALYs (95% CI: 4.43B DALYs-12.1B DALYs) lost**
- **\$1.19 quadrillion (95% CI: \$443T-\$2.41 quadrillion) economic destruction**
- **3.07k:1 (95% CI: 2.88k:1-3.12k:1) harm ratio** (Type II vs. Type I errors)

The 3.07k:1 (95% CI: 2.88k:1-3.12k:1) ratio demonstrates that these costs dwarf the benefits. The regulatory framework optimizes for bureaucratic risk minimization (avoiding blame for approvals) rather than population health maximization (saving lives).

The path forward is clear: maintain safety testing, eliminate efficacy delay, and deploy distributed real-world evidence systems.

⁴Institute of Medicine data shows 127.4 active physicians per 100,000 population in 1960. At US population of ~180M, this equals approximately 229,000 active physicians.

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*The NIH Pragmatic Trials Collaboratory funds trials at **\$500K for planning phase, \$1M/year. for implementation**—a tiny fraction of NIH’s budget. The ADAPTABLE trial cost **\$14 million** for **15,076 patients** (= **\$929/patient**) versus **\$420 million** for a similar traditional RCT (30x cheaper), yet pragmatic trials remain severely underfunded. PCORnet infrastructure enables real-world trials embedded in healthcare systems, but receives minimal support compared to basic research funding. Additional sources: <https://commonfund.nih.gov/hcscollaboratory> | https://pcornet.org/wp-content/uploads/2025/08/ADAPTABLE_Lay_Summary_21JUL2025.pdf | <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5604499/>*
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Mean exclusion rate: 86.1% across 158 antidepressant efficacy trials (range: 44.4% to 99.8%) More than 82% of real-world depression patients would be ineligible for antidepressant registration trials Exclusion rates increased over time: 91.4% (2010-2014) vs. 83.8% (1995-2009) Most common exclusions: comorbid psychiatric disorders, age restrictions, insufficient depression severity, medical conditions Emergency psychiatry patients: only 3.3% eligible (96.7% excluded) when applying 9 common exclusion criteria Only a minority of depressed patients seen in clinical practice are likely to be eligible for most AETs Note: Generalizability of antidepressant trials has decreased over time, with increasingly stringent exclusion criteria eliminating patients who would actually use the drugs in clinical practice Additional sources: <https://pubmed.ncbi.nlm.nih.gov/26276679/> | <https://pubmed.ncbi.nlm.nih.gov/26164052/> | <https://www.wolterskluwer.com/en/news/antidepressant-trials-exclude-most-real-world-patients-with-depression>
3. CNBC. Warren buffett’s career average investment return. *CNBC* <https://www.cnbc.com/2025/05/05/warren-buffetts-return-tally-after-60-years-5502284percent.html> (2025)
Berkshire’s compounded annual return from 1965 through 2024 was 19.9%, nearly double the 10.4% recorded by the S&P 500. Berkshire shares skyrocketed 5,502,284% compared to the S&P 500’s 39,054% rise during that period. Additional sources: <https://www.cnbc.com/2025/05/05/warren-buffetts-return-tally-after-60-years-5502284percent.html> | <https://www.slickcharts.com/berkshire-hathaway/returns>
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US agricultural subsidies total approximately \$30 billion annually, but create much larger economic distortions. Top 10% of farms receive 78% of subsidies, benefits concentrated in commodity crops (corn, soy, wheat, cotton), environmental damage from monoculture incentivized, and overall deadweight loss estimated at \$50-120 billion annually. Additional sources: <https://farm.ewg.org/> | <https://www.ers.usda.gov/topics/farm-economy/farm-sector-income-finances/government-payments-the-safety-net/>
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Since 1971, the war on drugs has cost the United States an estimated \$1 trillion in enforcement. The federal drug control budget was \$41 billion in 2022. Mass incarceration costs the U.S. at least \$182 billion every year, with over \$450 billion spent to incarcerate individuals on drug charges in federal prisons.
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General range: \$3,000-\$5,500 per life saved (GiveWell top charities) Helen Keller International. (Vitamin A): \$3,500 average (2022-2024); varies \$1,000-\$8,500 by country Against Malaria Foundation: \$5,500 per life saved New Incentives (vaccination incentives): \$4,500 per life saved Malaria Consortium (seasonal malaria chemoprevention): \$3,500 per life saved VAS program details: \$2 to provide vitamin A supplements to child for one year Note: Figures accurate for 2024. Helen Keller VAS program has wide country variation (\$1K-\$8.5K) but \$3,500 is accurate average. Among most cost-effective interventions globally Additional sources: <https://www.givewell.org/charities/top-charities> | <https://www.givewell.org/charities/helen-keller-international> | <https://ourworldindata.org/cost-effectiveness>
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Average family caregiver: 25-26 hours per week (100-104 hours per month) 38 million. caregivers providing 36 billion hours of care annually Economic value: \$16.59 per hour = \$600 billion total annual value (2021) 28% of people provided eldercare on a given day, averaging 3.9 hours when providing care Caregivers living with care recipient: 37.4 hours per week Caregivers not living with recipient: 23.7 hours per week Note: Disease-related caregiving is subset of total; includes elderly care, disability care, and child care Additional sources: <https://www.aarp.org/caregiving/financial-legal/info-2023/unpaid-caregivers-provide-billions-in-care.html> | <https://www.bls.gov/news.release/elcare.nr0.htm> | <https://www.caregiver.org/resource/caregiver-statistics-demographics/>
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As many as 80% of all heart attacks and strokes are preventable" through addressing lifestyle risk factors WHO: Most cardiovascular diseases can be prevented by addressing behavioral and environmental risk factors such as tobacco use, unhealthy diet, obesity, physical inactivity, harmful use of alcohol, and air pollution Cleveland Clinic: "90 percent of heart disease is preventable through healthier diet, regular exercise, and not smoking In 2 large cohort studies, a reduction of CVD risk of >80% and diabetes >90% were demonstrated in individuals who followed healthy lifestyle practices Only 5% of individuals follow all lifestyle factors for "ideal" cardiovascular health (American Heart Association) Additional sources: [https://www.who.int/news-room/fact-sheets/detail/cardiovascular-diseases-\(cvds\)](https://www.who.int/news-room/fact-sheets/detail/cardiovascular-diseases-(cvds)) | <https://world-heart-federation.org/what-we-do/prevention/> | <https://newsroom.cleveland-clinic.org/2021/09/29/90-percent-of-heart-disease-is-preventable-through-healthier-diet-regular-exercise-and-not-smoking> | <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC162259/>

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R3 paradigm: Rejuvenation (restoring cell function), Regeneration (stimulating repair), Replacement (substituting lost cells) Alzheimer's: Stanford study showed stem cell transplants reduced brain abnormalities in mice; Neural Stem Cell Therapy shown to improve cognitive function and reduce amyloid plaques Parkinson's: Clinical trial found Neural Stem Cell Therapy significantly improved motor function and was well-tolerated ALS/Huntington's: MSC (mesenchymal stem cell) therapy effectiveness confirmed; slows ALS progression Challenge: Clinical trials often enroll patients at advanced stages; many preclinically promising drugs ineffective in late-stage human trials Current status: Largely experimental and early clinical trial stages; researchers working to overcome delivery challenges, safety concerns, and targeting widespread neuronal damage Note: 80% theoretical max for neurodegenerative reflects early-stage intervention potential; current treatments limited (10% effective) Additional sources: <https://pmc.ncbi.nlm.nih.gov/articles/PMC7926761/> | <https://med.stanford.edu/news/all-news/2023/09/stem-cell-alzheimers.html> | <https://stemcellres.biomedcentral.com/articles/10.1186/s13287-025-04285-7>
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Phase III trial of drisapersen for Duchenne muscular dystrophy failed to show clinical benefit (P=0.415). FDA rejected approval in 2016 due to safety concerns including thrombocytopenia in 2% of patients, renal toxicity, and injection site reactions in 79% of patients. BioMarin discontinued development. Example of regulatory system preventing approval of ineffective drug with serious adverse effects.
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Graham testimony (2004): 88,000-139,000 U.S. heart attacks/strokes from Vioxx; up to 55,000 deaths (40% fatality rate) Lancet study estimate: 88,000 Americans had heart attacks from Vioxx; 38,000 died FDA memo (2004): Vioxx contributed to 27,785 heart attacks and sudden cardiac deaths (1999-2003) High-dose Vioxx: Tripled risk of heart attacks and sudden cardiac death Prescriptions: 92.8 million U.S. prescriptions 1999-2003 Withdrawn: September 30, 2004 after APPROVE trial showed cardiovascular risks Note: Vioxx case demonstrates failure of passive post-market surveillance (FAERS) to detect safety signals in time. Voluntary reporting missed cardiovascular risks for years despite millions of prescriptions Additional sources: https://pmc.ncbi.nlm.nih.gov/articles/PMC534432/ | https://www.npr.org/2007/11/10/5470430/timeline-the-rise-and-fall-of-vioxx | https://www.thelancet.com/journals/lancet/article/PIIS0140-6736(05
154. FDLI. Japan's regenerative medicine act and conditional approval pathway. *FDLI: Japan's Regen Med Pathways* <https://www.fdpi.org/2019/02/global-focus-japans-regenerative-medicine-regulatory-pathways-encouraging-innovation-and-patient-access/> (2019)
Act on Safety of Regenerative Medicine (RM Act) + amended Pharmaceuticals and Medical Devices Act (PMD Act): passed Nov 2013, effective Nov 2014 Conditional and time-limited approval pathway: Obtain approval after exploratory trials demonstrate probable benefit and proven safety 7-year conditional approval period to confirm clinical benefit (e.g., using surrogate endpoints) SAKIGAKE designation (April 2015): Expedited pathway for innovative products targeting serious/life-threatening diseases without effective treatment Benefits: Prioritized consultation, accelerated review, extended re-examination period, premium pricing Examples: Terumo's HeartSheet and Stemirac obtained conditional approval; Stemirac also SAKIGAKE-designated Additional sources: https://www.fdpi.org/2019/02/global-focus-japans-regenerative-medicine-regulatory-pathways-encouraging-innovation-and-patient-access/ | https://www.insights.bio/cell-and-gene-therapy-insights/journal/article/310/Experiences-from-Japan-SAKIGAKE-Designation-System-for-Regenerative-Medical-Products | https://pmc.ncbi.nlm.nih.gov/articles/PMC6696404/
155. Honmou, O. *et al.* [Intravenous infusion of auto serum-expanded autologous mesenchymal stem cells in spinal cord injury patients: 13 case series](#). *Clinical Neurology and Neurosurgery* **203**, 106565 (2021)
In this investigator-initiated clinical trial (2014-2017), 12 of 13 patients (92%) with cervical spinal cord injury who received MSC transplantation achieved at least one level of improvement on American Spinal Injury Association Impairment Scale. Five of six AIS A patients improved to AIS B (3/6) or AIS C (2/6). No severe side effects. Results led to MHLW conditional approval of STEMIRAC in December 2018.

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Reviews CAR-T cell therapy access across Asia. Japan has approved five CAR-T products under National Health Insurance: Kymriah, Yescarta, Brexanzi, Abecma, and Carvykti. All priced at 32,647,761 JPY. Japan's high-cost medical care reimbursement system limits patient out-of-pocket costs.
157. Medicine (US) Committee on the U. S. Physician Supply, I. of. U.s. Physician supply and requirements: Match or mismatch? *The Nation's Physician Workforce: Options for Balancing Supply and Requirements* <https://www.ncbi.nlm.nih.gov/books/NBK232541/> (1996)
Historical physician workforce data: 126.6 active nonfederal M.D. physicians per 100,000 population in 1950; 127.4 in 1960; 137.4 in 1970 (DHHS 1993). At 1960 US population of 180M, this corresponds to approximately 229,000 active physicians.